

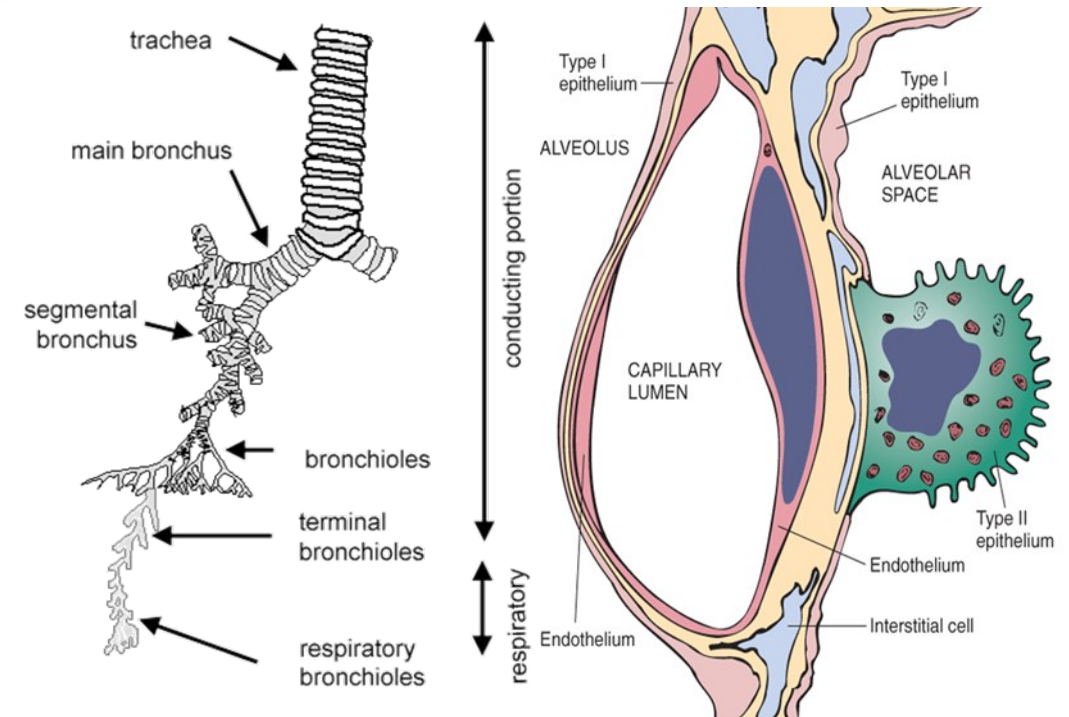
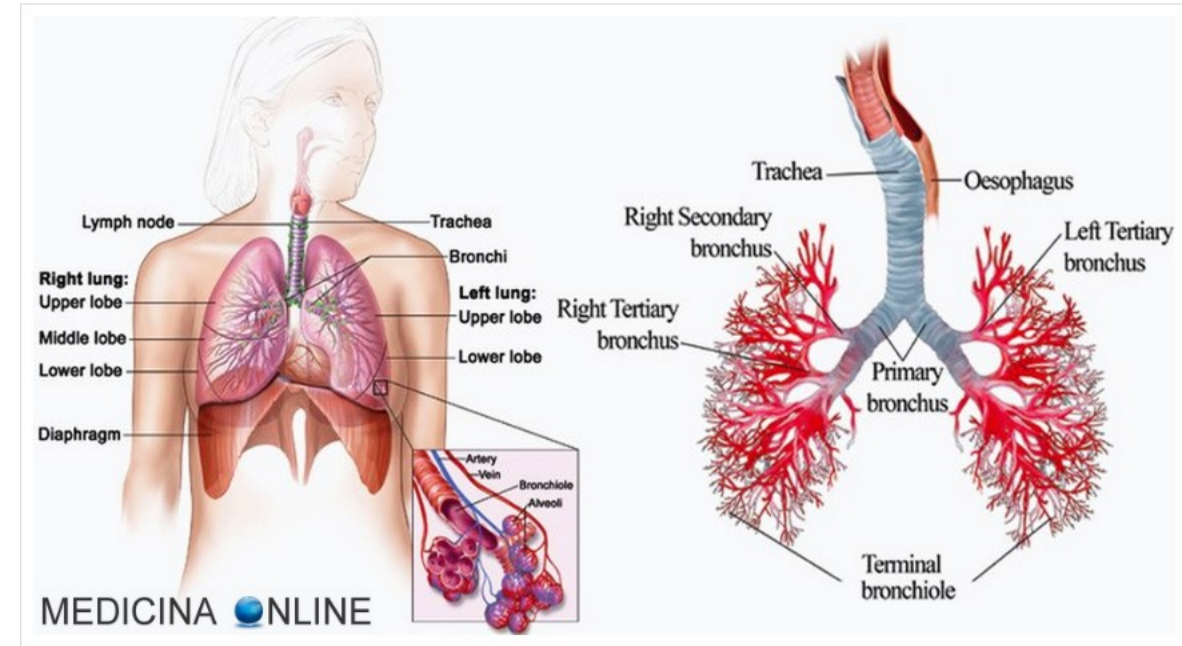
Diseases of the lower respiratory tract

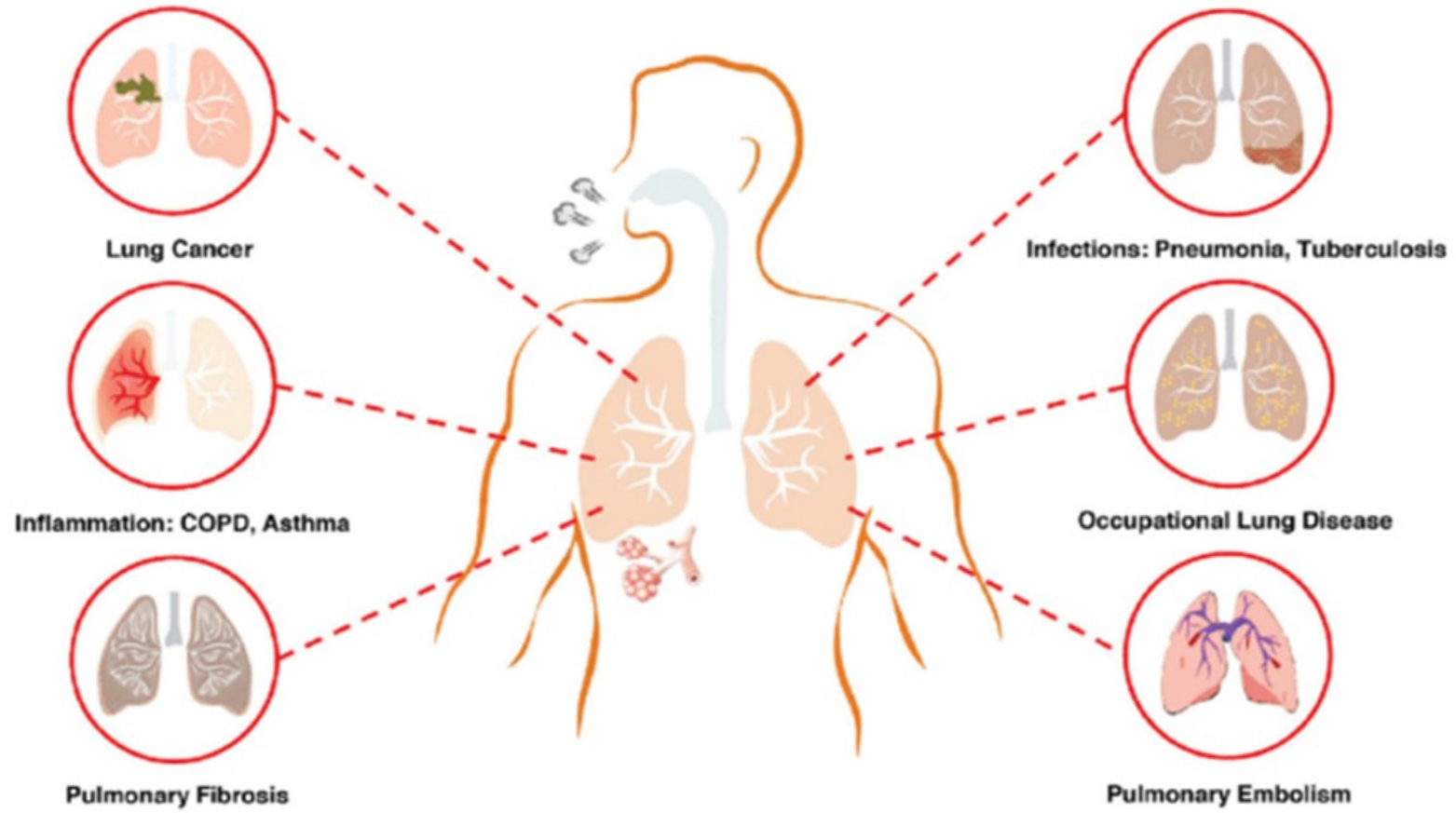
BY MELESE A.(MD, PATHOLOGIST)

Introduction

Anatomy:

- ❖ The lower respiratory system consists of the:
 - Trachea
 - Bronchi and Bronchioles (the conducting airways)
 - Alveoli (the gas-exchange units)
 - Pleura (the lining of the lungs and chest cavity)
- ❖ **Bronchi** divide dichotomously to give **terminal** bronchioles → **Acinus**(respiratory bronchioles) → **alveolar ducts** → **alveolar sacs**(blind ends of respiratory passages)
- ❖ Function of the lung: **Oxygenation of blood** and **elimination of carbon dioxide**.

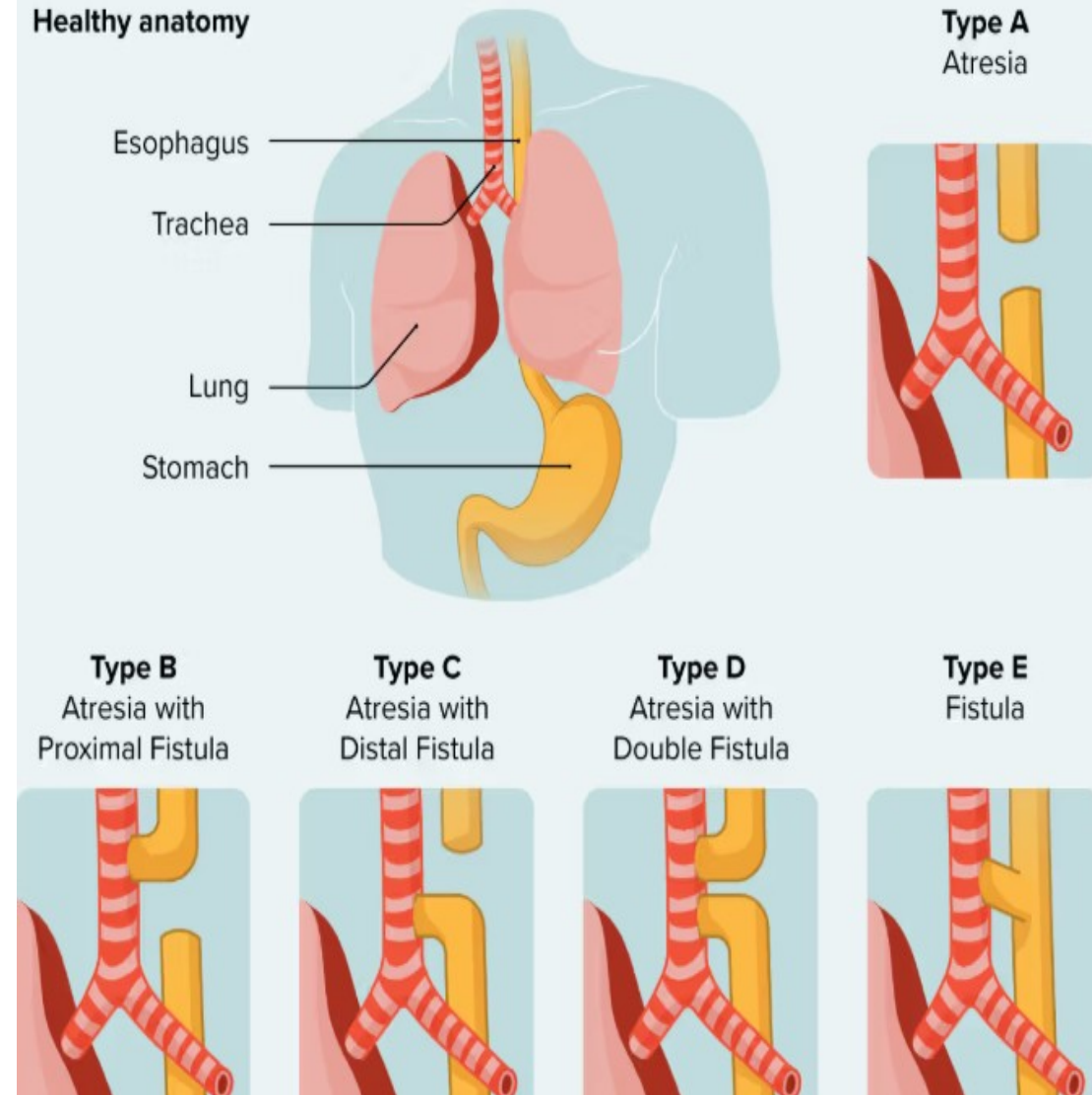




Congenital Anomalies

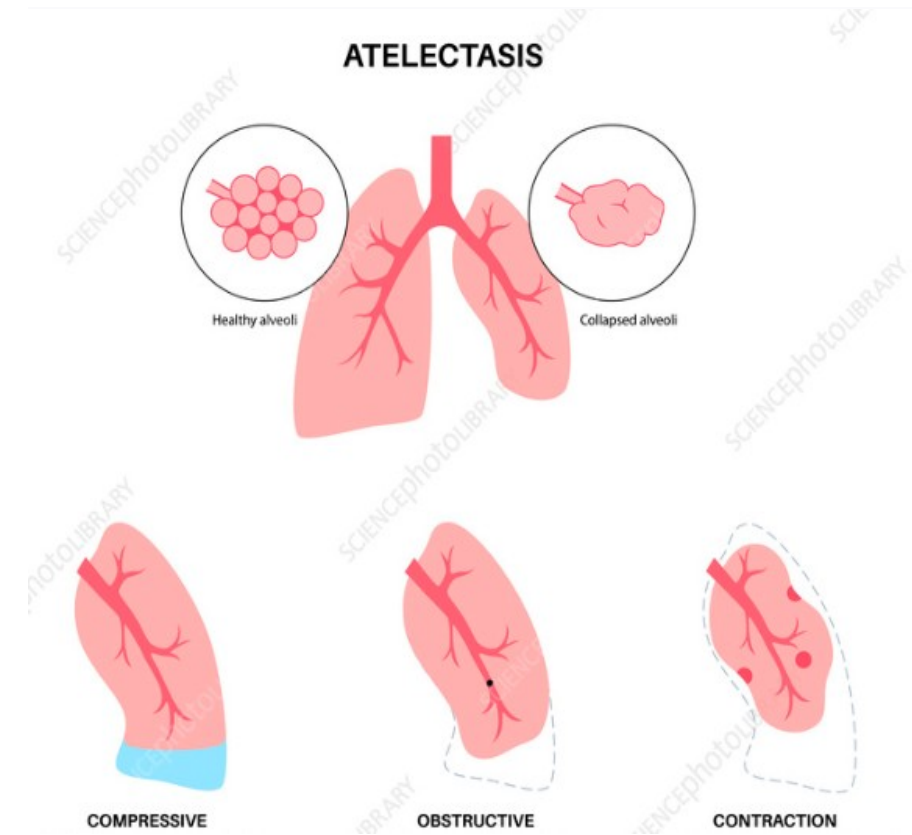
- ❖ Pulmonary hypoplasia
- ❖ Foregut cysts
- ❖ Pulmonary sequestration: a discrete area of lung tissue
 - Lacks any connection to the airway system
 - Has an **abnormal blood supply** arising from the **aorta** or its branches.
 - Extralobar sequestrations are **external** to the lung
 - Intralobar sequestrations occur **within** the lung.
- ❖ Tracheoesophageal Fistula (TEF) & Esophageal atresia
 - Anomalies involving an **abnormal connection (fistula)** between the **trachea and esophagus**, often with a **blind-ending pouch in the esophagus (atresia)**.
 - The most common type
 - **blind proximal esophagus** with a fistula from the **distal esophagus to the trachea**.

Types of Esophageal Atresia and Tracheoesophageal Fistula



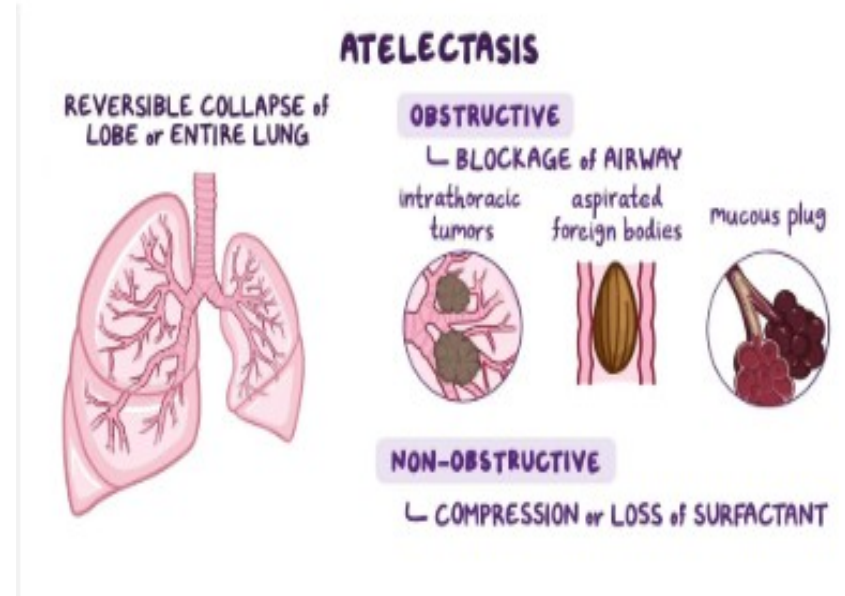
Atelectasis (Collapse)

- ❖ Atelectasis refers either to
 - **Incomplete expansion** of the lungs (**neonatal atelectasis**)
 - **Collapse of previously inflated lung**, producing areas of relatively **airless** pulmonary parenchyma.
- ❖ This results in shunting of blood through non-ventilated areas, leading to **impaired gas exchange**.
- ❖ Atelectasis is classified into **three forms**
 - **Resorption**
 - **Compression**
 - **Contraction**



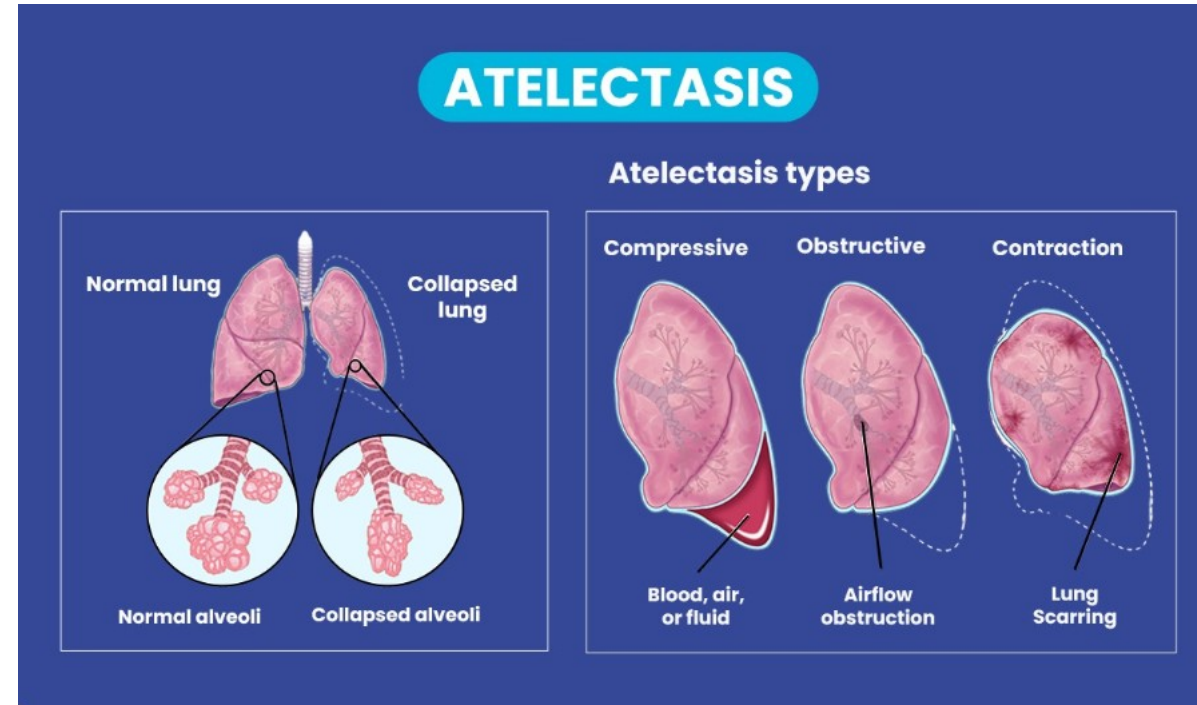
Resorption Atelectasis

- ❖ Stems from **complete obstruction** of an airway.
- ❖ Occurs when an **obstruction prevents** air from reaching distal airways.
- ❖ The air already present gradually becomes **absorbed**, and **alveolar collapse** follows
- ❖ Depending on the **level of airway obstruction**, an **entire lung**, a **complete lobe**, or one or more **segments** may be involved
- ❖ Common cause is obstruction of a bronchus by
 - mucous or mucopurulent plug
 - Aspiration of foreign bodies
 - fragments of bronchial tumors
- ❖ Frequently occurs **postoperatively**
- ❖ The **mediastinum shifts toward** the atelectatic lung.



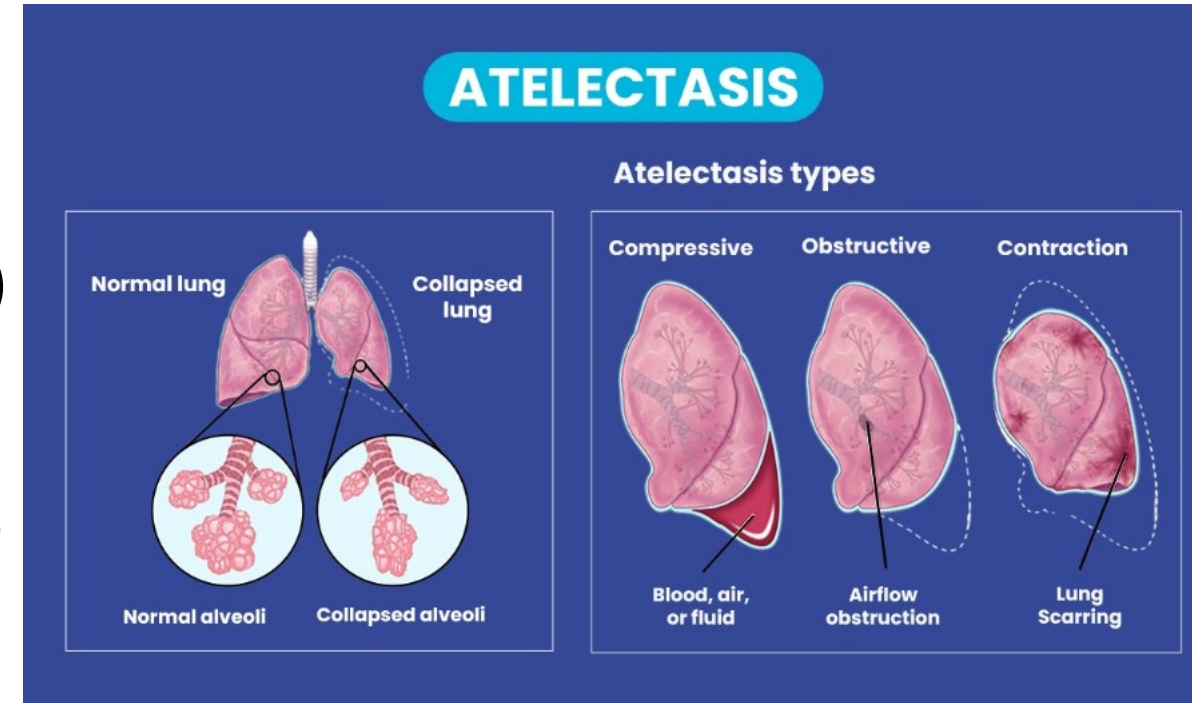
Compression Atelectasis

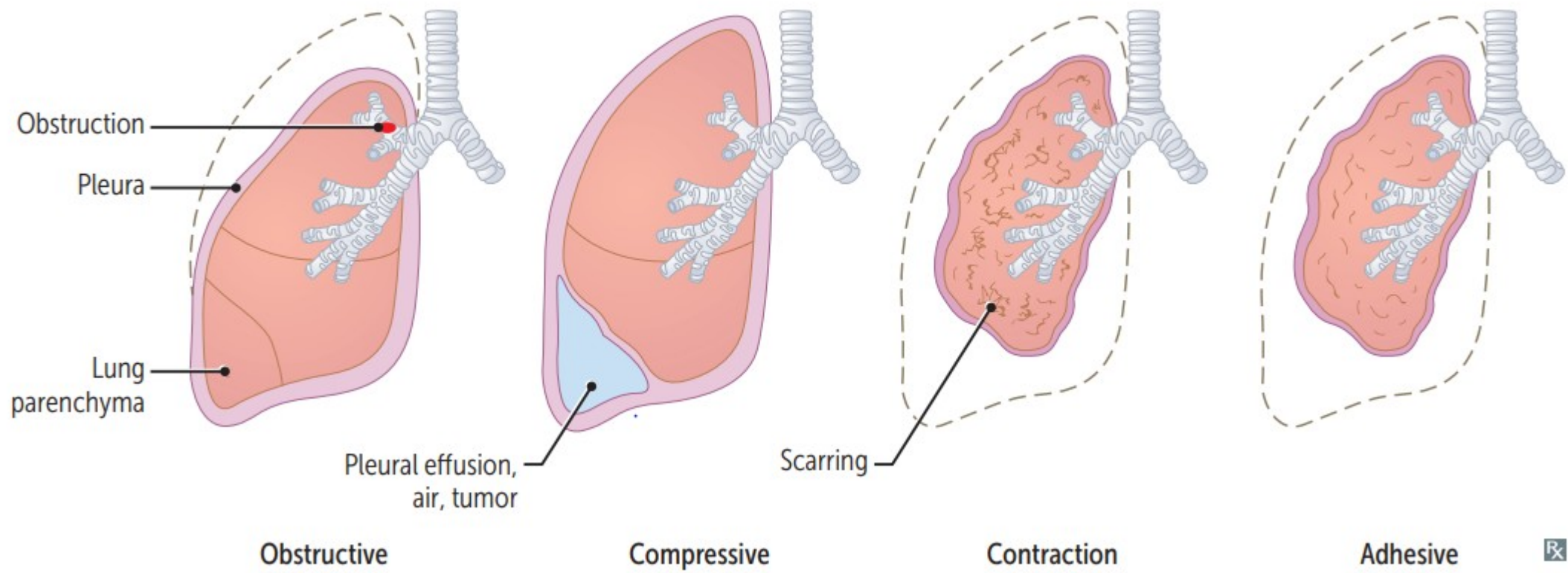
- ❖ Called **passive** or **relaxation** atelectasis
- ❖ Is usually associated with **accumulations of fluid, blood, or air** within the **pleural cavity**
- ❖ **Basal atelectasis** resulting from the **elevated position of the diaphragm** commonly occurs
 - In bedridden patients
 - in patients with ascites
 - in patients **during** and **after** surgery
- ❖ The **mediastinum** shifts ***away from*** the affected lung.



Contraction Atelectasis

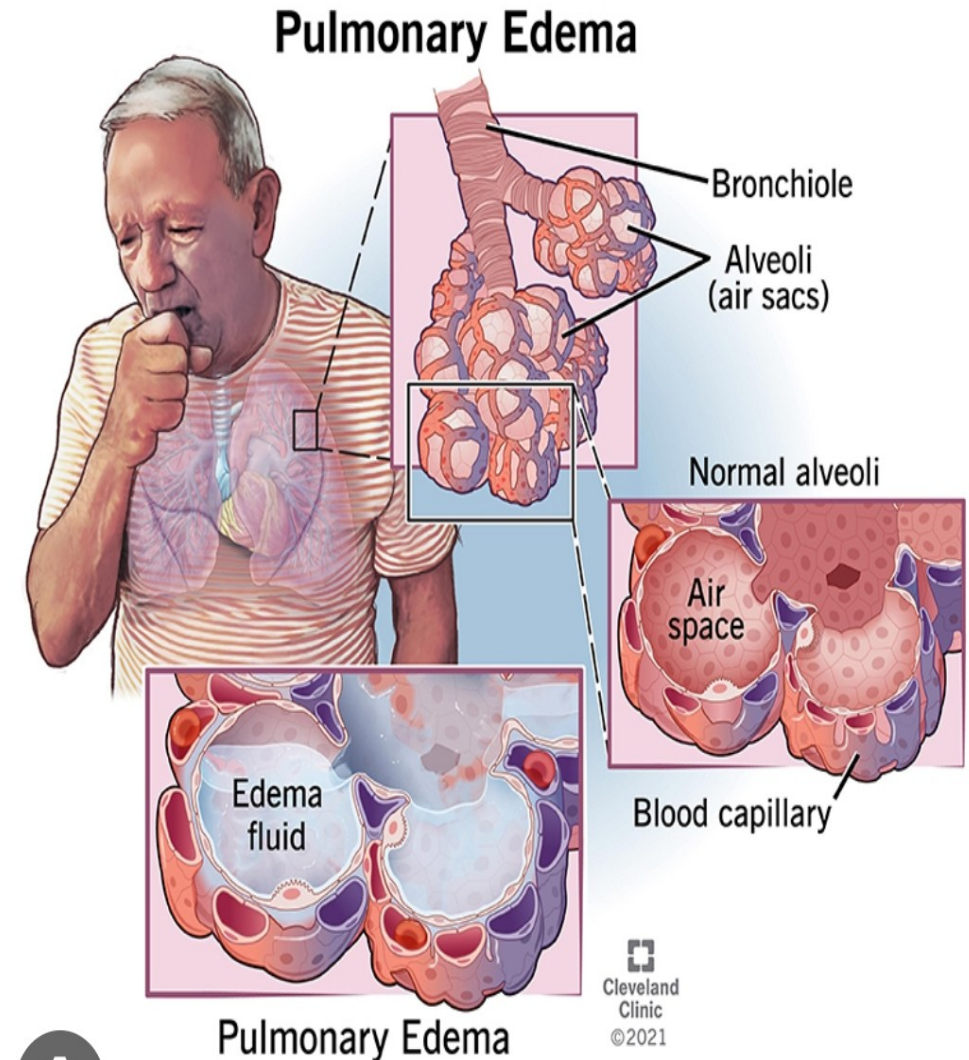
- ❖ Occurs when **focal** or **generalized** pulmonary or pleural **fibrosis** prevents **full lung expansion**.
- ❖ Atelectasis (except that caused by contraction) is potentially reversible
- ❖ Should be treated promptly to **prevent hypoxemia** and **superimposed infection** of the collapsed lung





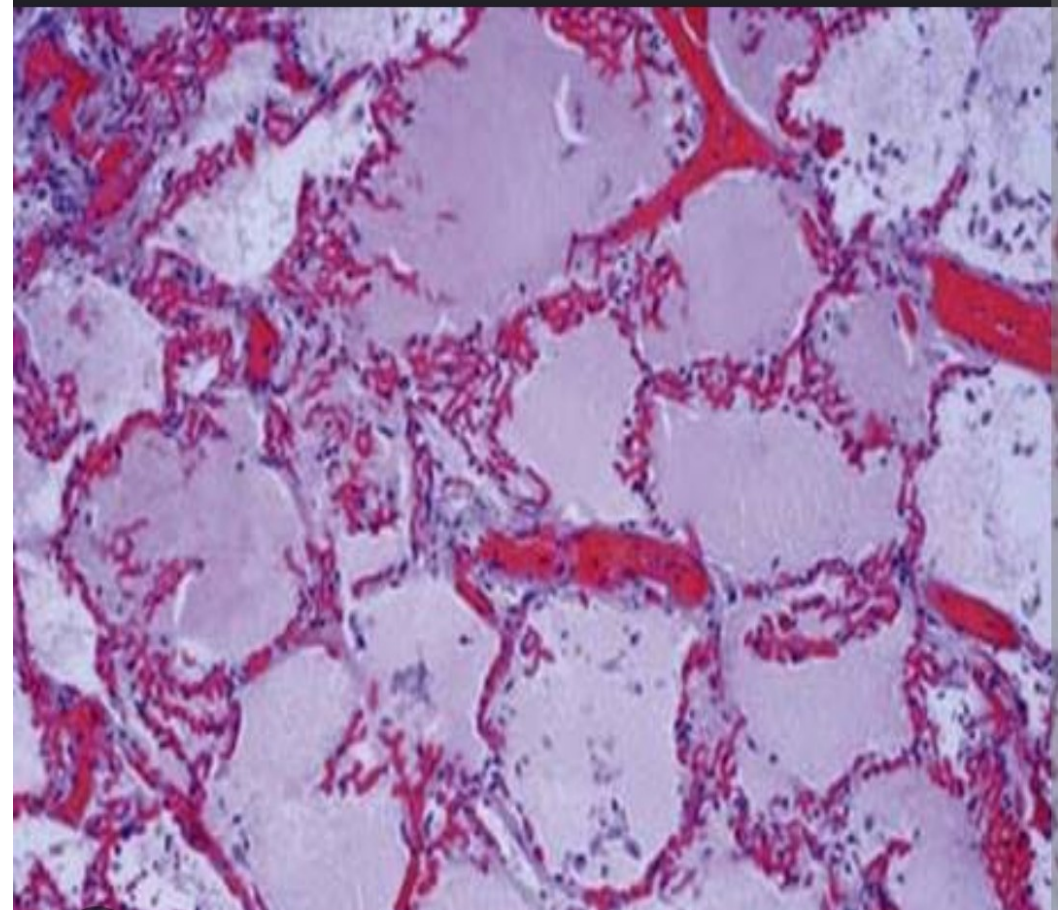
Pulmonary Edema

- ❖ Abnormal accumulation of fluid in the lung parenchyma, primarily within the **alveoli** and **interstitial spaces**.
- ❖ Can result from
 - Hemodynamic disturbances (cardiogenic pulmonary edema)
 - Direct increases in capillary permeability (microvascular injury)



Hemodynamic Pulmonary Edema

- ❖ Is due to **increased hydrostatic pressure**
 - occurs most commonly in **left sided congestive heart failure**.
- ❖ Fluid accumulates initially in the **basal regions** of the lower lobes because hydrostatic pressure is greatest in these sites (**dependent** edema).
- ❖ Histologically
 - The alveolar **capillaries** are engorged
 - **Intra-alveolar transudate** appears as ***finely granular pale pink*** material.
- ❖ Alveolar microhemorrhages and hemosiderin laden macrophages (“heart failure” cells) may be present.



Edema Caused by Microvascular (Alveolar) Injury

- ❖ Non cardiogenic pulmonary edema is due to **injury to the alveolar septa**.
- ❖ Primary injury to the **vascular endothelium** or **damage to alveolar epithelial cells** (with secondary microvascular injury)

Pathogenesis:

- **Direct Injury** (e.g., pneumonia, aspiration, smoke inhalation) or **Indirect Injury** (e.g., sepsis, trauma, pancreatitis) → Activation of inflammatory mediators (cytokines, neutrophils) → Damage to capillary endothelium and alveolar epithelium → Increased Permeability.
- ❖ Fluid, proteins, and even red blood cells leak freely into the alveolar spaces. The lymphatic system is overwhelmed.

Acute Lung Injury and Acute Respiratory Distress Syndrome (Diffuse Alveolar Damage)

- ❖ **Acute lung injury (ALI)** (also called **noncardiogenic pulmonary edema**)
 - Is characterized by the **abrupt onset** of *significant hypoxemia* and *bilateral pulmonary infiltrates* in the **absence of cardiac failure**.
- ❖ **Acute respiratory distress syndrome (ARDS)** is a manifestation of **severe ALI**.
- ❖ **ARDS** is a clinical syndrome of **progressive respiratory insufficiency** caused by **diffuse alveolar damage** in the setting of
 - sepsis, severe trauma, or diffuse pulmonary infection
- ❖ Both ARDS and ALI are associated with
 - Increases in **pulmonary vascular permeability**, edema and epithelial cell death.
- ❖ The **histologic** manifestation of these diseases is ***diffuse alveolar damage (DAD)***.
 - hyaline membranes (a characteristic feature of ALI/ARDS)

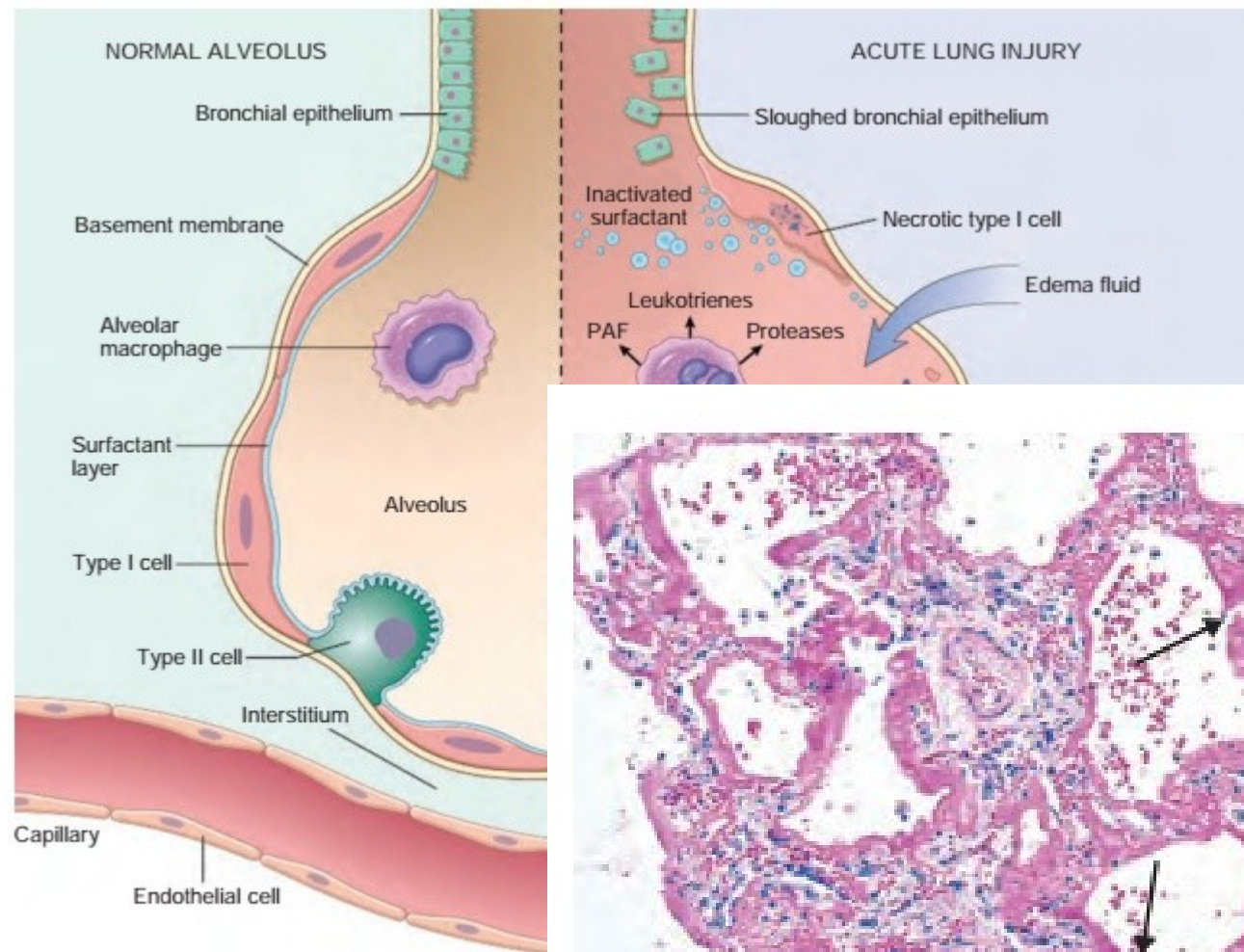


Figure 15-3 The normal alveolus (*left side*) compared with the injured (Modified with permission from Matthay MA, Ware LB, Zimmerman leukin-1; *MIF*, migration inhibitory factor; *PAF*, platelet activating fact

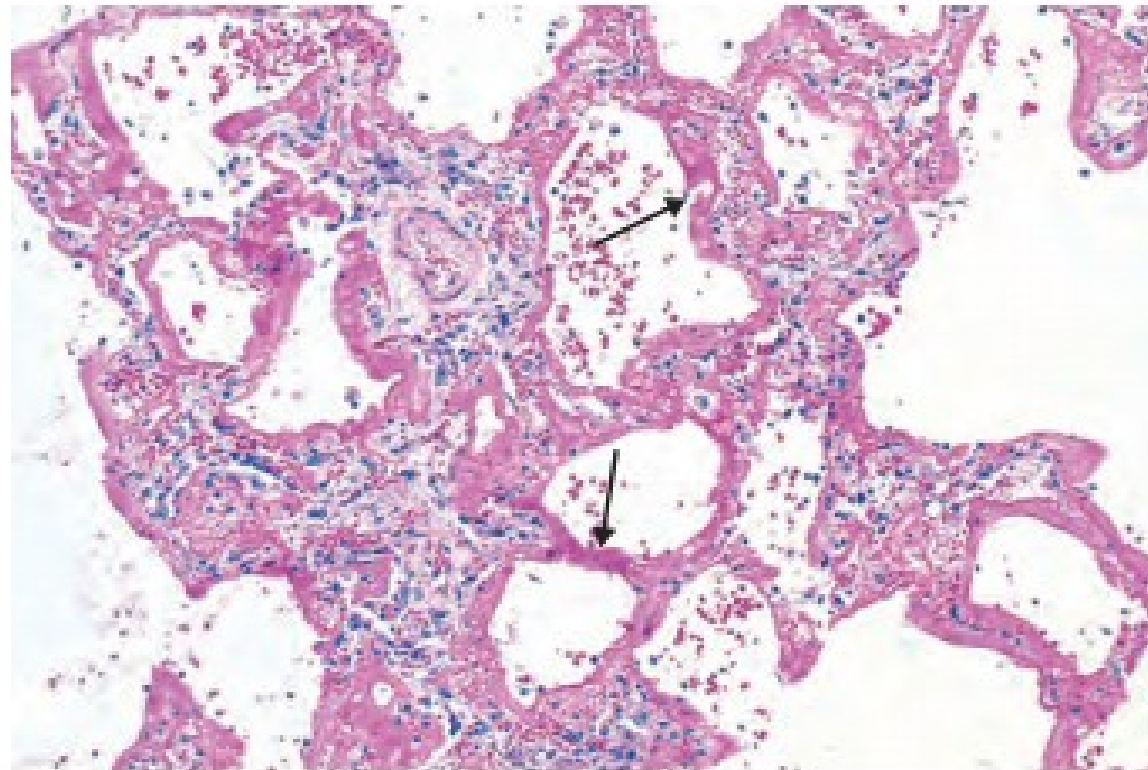


Figure 15-4 Diffuse alveolar damage (acute respiratory distress syndrome). Some of the alveoli are collapsed, while others are distended. Many are lined by hyaline membranes (*arrows*).

- ❖ Profound dyspnea & tachypnea herald ARDS
- ❖ Cyanosis, hypoxemia, resp.failure
- ❖ Hypoxemia becomes unresponsive to oxygen therapy
- ❖ Respiratory acidosis
- ❖ Mortality rate is about 60%
- ❖ X- ray may be normal initially, later there will be diffuse bilateral infiltrates

Table 15.2 Conditions Associated With Development of Acute Respiratory Distress Syndrome

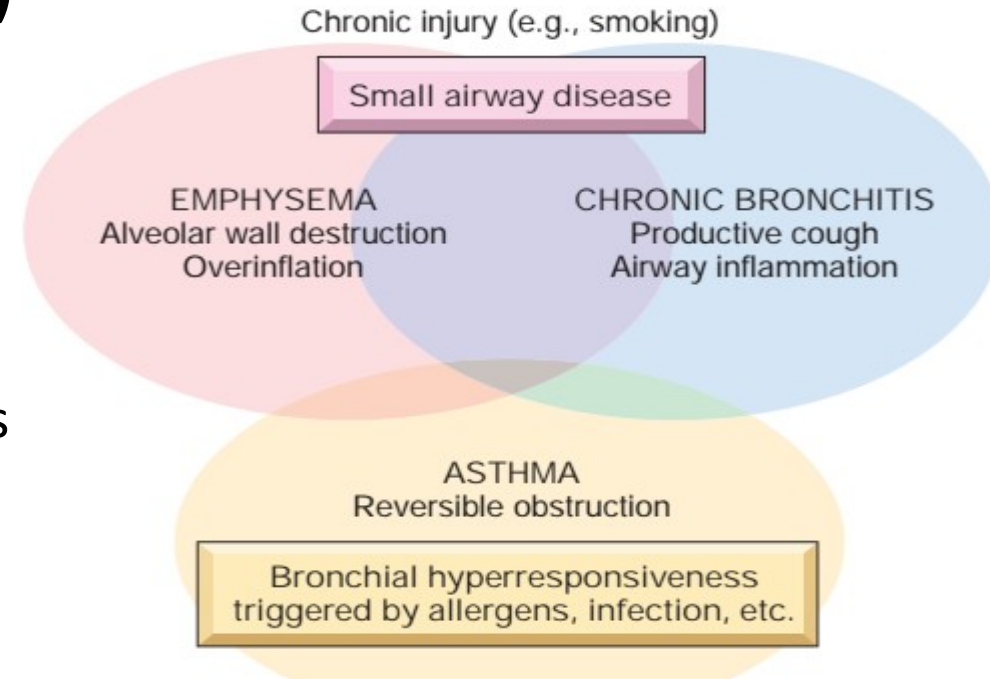
Infection
Sepsis ^a Diffuse pulmonary infections ^a Viral, <i>Mycoplasma</i> , and <i>Pneumocystis</i> pneumonia; miliary tuberculosis Gastric aspiration ^a
Physical/Injury
Mechanical trauma including head injuries ^a Pulmonary contusions Near-drowning Fractures with fat embolism Burns Ionizing radiation
Inhaled Irritants
Oxygen toxicity Smoke Irritant gases and chemicals
Chemical Injury
Heroin or methadone overdose Acetylsalicylic acid Barbiturate overdose Paraquat
Hematologic Conditions
Transfusion-associated lung injury (TRALI) Disseminated intravascular coagulation
Pancreatitis
Uremia
Cardiopulmonary Bypass
Hypersensitivity Reactions
Organic solvents Drugs

^aMore than 50% of cases of acute respiratory distress syndrome are associated with these four conditions.

Obstructive and Restrictive Lung Diseases

Obstructive lung diseases (airway diseases)

- ❖ Are characterized by an **increase in resistance** to airflow due to **partial or complete obstruction** at any level
- ❖ Pulmonary function tests
 - **decreased** maximal airflow rates during forced **expiration** => expressed as **FEV1/FVC**
 - An FEV1/FVC ratio of **less than 0.7** generally indicates **airway obstruction**
- ❖ The major diffuse obstructive disorders
 - Emphysema
 - Chronic bronchitis
 - Bronchiectasis
 - Asthma

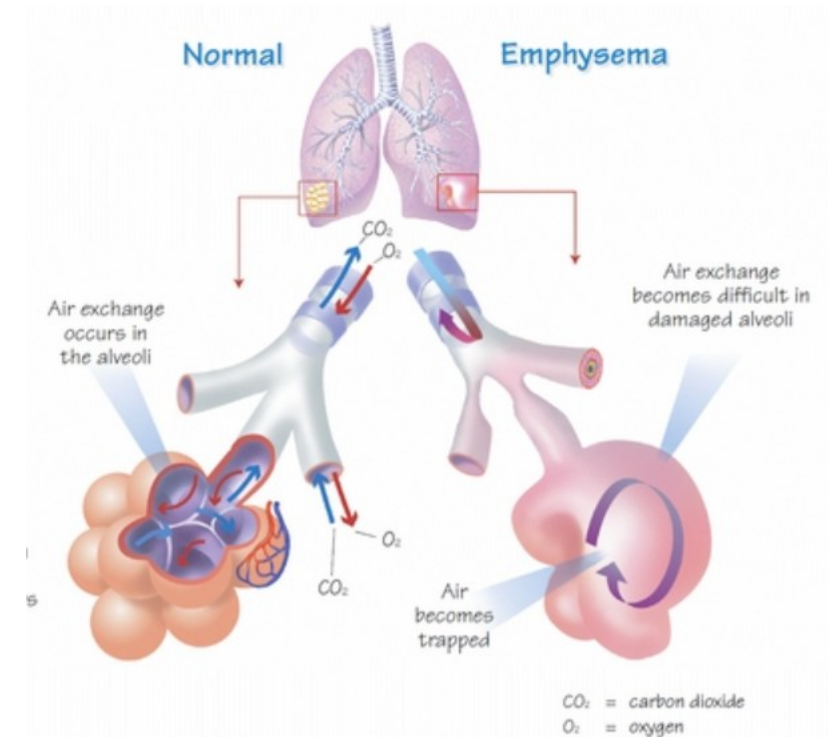


Restrictive diseases

- ❖ Are characterized by **reduced expansion** of lung parenchyma and **decreased total lung capacity**.
- ❖ **Proportionate decreases** in both **total lung capacity** and **FEV1**, leading to **normal FEV1/FVC ratio**.
- ❖ FVC is reduced and the **expiratory flow rate** is **normal**
- ❖ Two broad kinds of conditions:
 - **chest wall disorders** (e.g., severe obesity, pleural diseases, kyphoscoliosis, and neuromuscular diseases such as poliomyelitis)
 - **chronic interstitial and infiltrative diseases**, such as pneumoconioses and interstitial fibrosis.

Emphysema

- ❖ Is characterized by **irreversible enlargement** of the airspaces **distal to the terminal bronchiole**
 - accompanied by **destruction of their walls with subtle small air way fibrosis.**
- ❖ Emphysema is classified according to its ***anatomic distribution within the lobule***
- ❖ classified into four major types:
 - Centriacinar
 - Panacinar
 - Paraseptal
 - Irregular

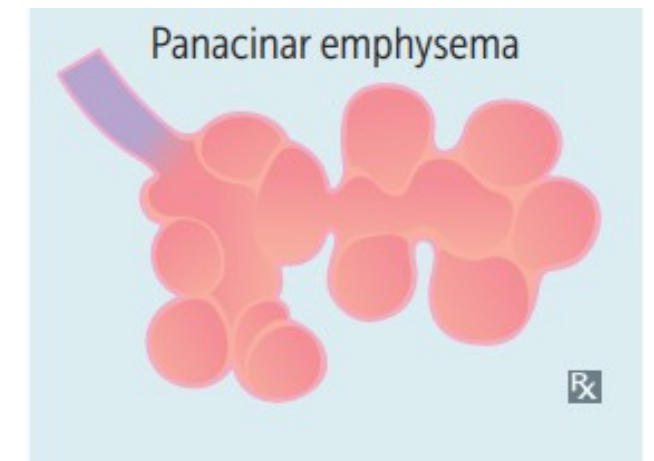
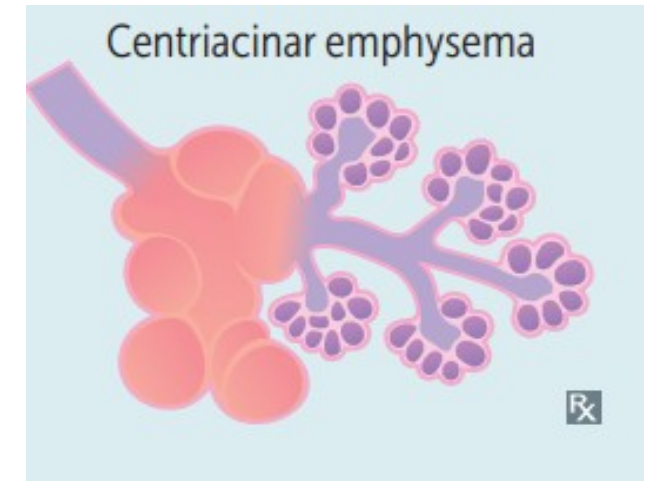


1. Centriacinar => The most common (>95%)

- Destruction of **central portion** with sparing of distal airways
- Both **emphysematous** and **normal** airspaces exist within the same acinus and lobule
- **Upper lobes** > lower
- Cause: **smoking**

2. Panacinar

- The acini are **uniformly enlarged**
- occur more commonly in the **lower zones** and in the anterior margins of the lung
- it is usually most severe at the **bases**
- is associated with ***α1-antitrypsin deficiency***



Emphysema

3. Distal Acinar (paraseptal) Emphysema

- ❖ The **proximal** portion of the acinus is normal, but the **distal** part is predominantly involved.
- ❖ More striking adjacent to the **pleura**
- ❖ Occurs adjacent to areas of **fibrosis**, **scarring** or **atelectasis**
- ❖ More **severe** in the **upper half** of the lungs
- ❖ Sometimes form cyst like structures
- ❖ Underlies many of the **spontaneous pneumothorax**

4. Airspace enlargement with fibrosis (Irregular)

- ❖ The acinus is irregularly involved
- ❖ Associated with scarring
- ❖ Asymptomatic & clinically **insignificant**

Pathogenesis

- ❖ Inhaled **cigarette smoke** and other noxious particles cause lung damage and inflammation => which results in
 - **parenchymal destruction (emphysema)**
 - **airway disease** (bronchiolitis and chronic bronchitis)
- ❖ **Factors that influence the development of emphysema** include
 1. **Inflammatory mediators and leukocytes**
 2. **Protease-antiprotease imbalance.**
 - In **patients** who develop **emphysema**
 - There is a relative **deficiency of protective antiproteases**
 3. **Oxidant-antioxidant imbalance**
 - Substances in tobacco smoke, alveolar damage, and inflammatory cells all **produce oxidants**
 - Results in **functional α 1- AT deficiency**
 4. **Infection**

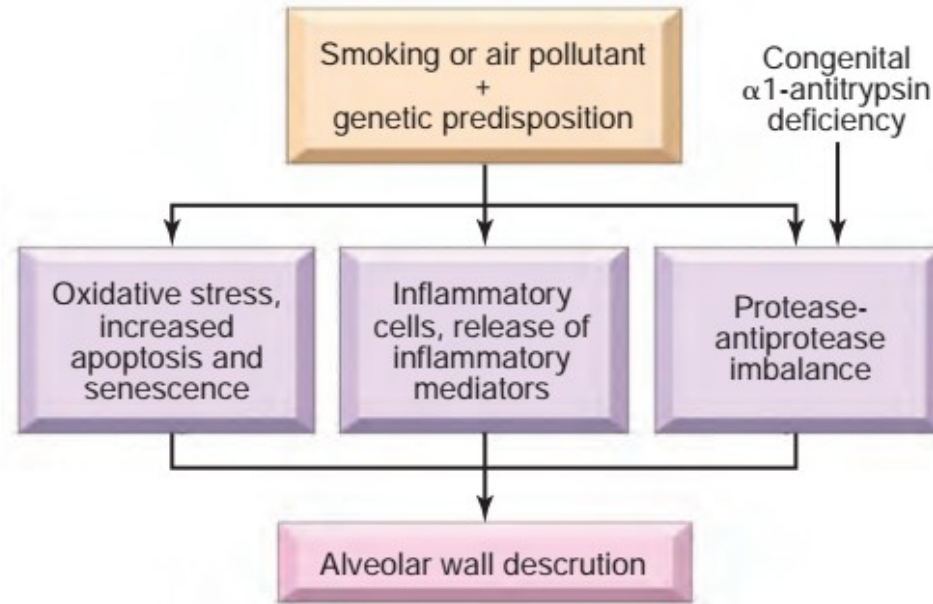
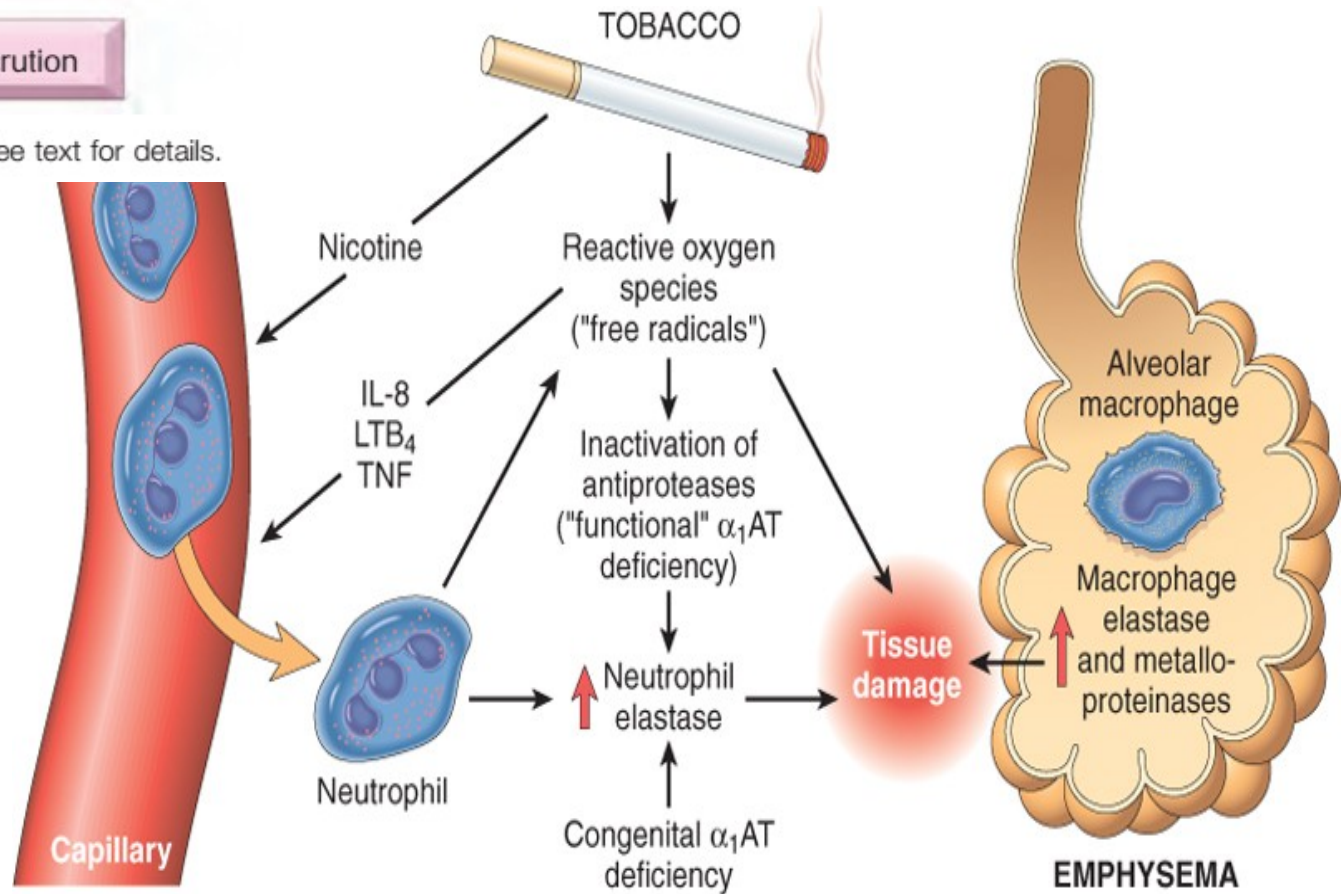


Figure 15-8 Pathogenesis of emphysema. See text for details.



MORPHOLOGY

- ❖ Advanced emphysema produces **voluminous lungs**
- ❖ Generally, the **upper two thirds** of the lungs are more severely affected.
- ❖ Large apical blebs or bullae are more characteristic of
 - **irregular emphysema** secondary to scarring
 - **distal acinar emphysema**.
- ❖ Microscopically
 - abnormally large alveoli are separated by thin septa with only focal centriacinar fibrosis.
- ❖ **Prolonged vasoconstriction** leads to changes of **pulmonary arterial HTN**.



Figure 15-9 Bullous emphysema. Note the large subpleural bullae (upper left).

Clinical Course

- ❖ Symptoms do not appear **until** at **least one third** of the functioning pulmonary parenchyma is damaged.
- ❖ **Dyspnea** usually appears first, beginning **insidiously** but **progressing steadily**.
- ❖ In some patients, **cough** or **wheezing** is the chief complaint, easily confused with asthma.
- ❖ Cough and expectoration
- ❖ **Weight loss** is common and can be so severe as to suggest an **occult cancer**.
- ❖ Classically, the patient with **severe emphysema** is **barrel-chested** and **dyspneic**
 - **sits forward in a hunched-over** position, and breathes through **pursed lips**
- ❖ Such patients may **over ventilate** and remain well oxygenated (***pink puffers***)

EMPHYSEMA

- “Barrel chest” from lung over-expansion
- Pursed-lip breathing
- Use of accessory muscles
- Tachypnoea
- Weak & Thin
- Decreased lung sounds



Clinical Course

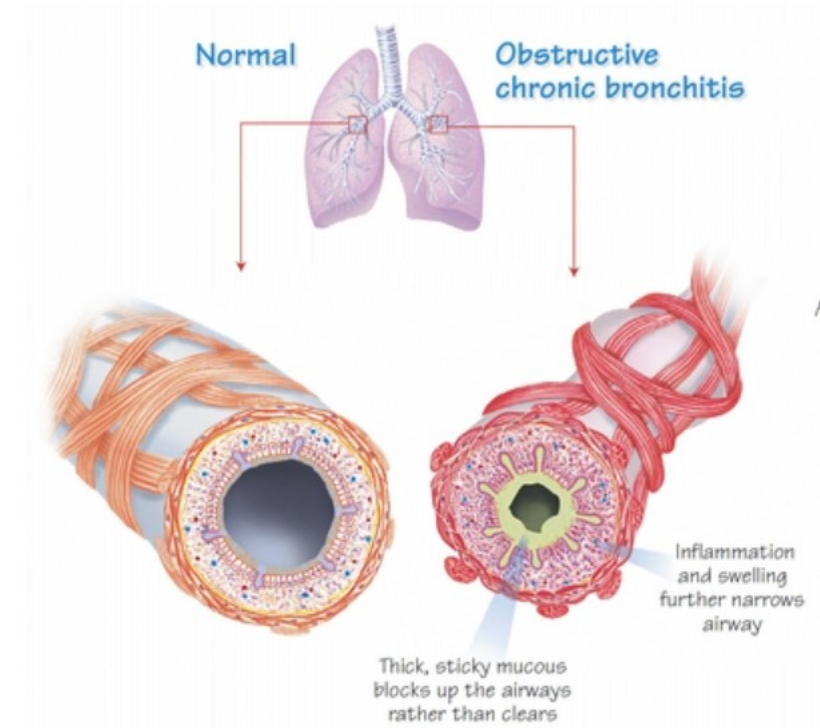
- ❖ Development of **cor pulmonale** and eventually **congestive heart failure**, related to **secondary pulmonary hypertension**, is associated with a **poor prognosis**.
- ❖ Death in most patients with emphysema is due to
 - coronary artery disease
 - respiratory failure
 - right sided heart failure
 - massive collapse of the lungs secondary to **pneumothorax**
- ❖ Treatment options include
 - smoking cessation
 - oxygen therapy
 - long-acting bronchodilators with inhaled corticosteroids

Chronic Bronchitis

- ❖ Is defined clinically as
 - **persistent cough** with **sputum production** for **at least 3 months** in **at least 2 consecutive years**, in the **absence** of any other identifiable **cause**.
- ❖ Common in **habitual smokers** and inhabitants of **smog-laden cities**
- ❖ When chronic bronchitis persists for years, lead to
 - Cor pulmonale and heart failure
 - Cause metaplasia and dysplasia => providing a **rich soil** for **cancerous transformation**.

Pathogenesis

- ❖ The **primary or initiating factor** in the genesis of chronic bronchitis is exposure to
 - **Tobacco smoke (90%)**
 - **Dust** from grain, cotton, and silica.
- ❖ **Mucus hypersecretion**
 - Earliest feature
 - Associated with hypertrophy of the submucosal glands
- ❖ **Inflammation**
 - Long-standing inflammation and accompanying **fibrosis** can also lead to **chronic airway obstruction**.
- ❖ **Infection**
 - Significant in maintaining it and may be critical in producing **acute exacerbations**.

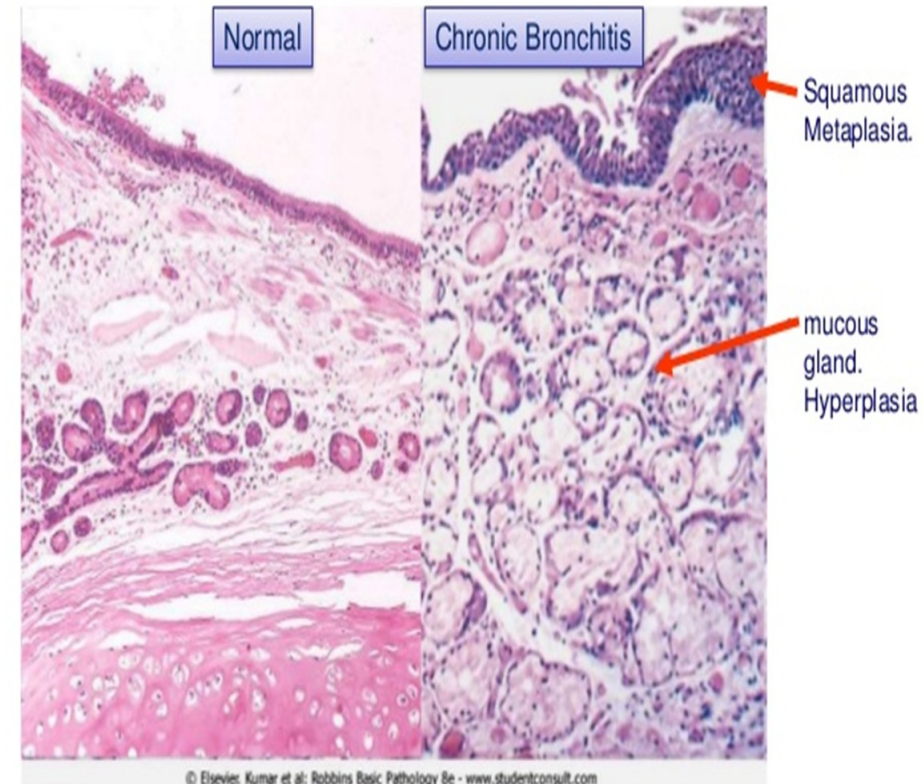


MORPHOLOGY

- ❖ Grossly, there is **hyperemia**, **swelling**, and **edema** of the **mucous membranes**
- ❖ The characteristic features are
 - **Mild chronic inflammation** of the airways
 - Enlargement of the **mucus-secreting glands**
 - Assessed by the **ratio** of the **thickness** of the **mucous gland layer** to the **thickness of the wall** between the epithelium and the cartilage (**Reid index**).
 - The Reid index (normally 0.4) is **increased** in chronic bronchitis
- ❖ The bronchial epithelium may exhibit **squamous metaplasia and dysplasia**.
- ❖ There is **marked narrowing of bronchioles** caused by
 - ❖ mucus plugging, inflammation, and fibrosis.
- ❖ In the most **severe cases**, there may be **obliteration** of lumen **due to fibrosis (bronchiolitis obliterans)**.



Chronic Bronchitis:



Clinical Features

- ❖ The cardinal symptom of chronic bronchitis is a **persistent cough productive of sputum**
- ❖ For many years **no other** respiratory functional impairment is present
- ❖ But eventually **dyspnea on exertion** develops.
- ❖ With the **passage of time**, and usually with **continued smoking**,
 - Hypercapnia, hypoxemia, and mild **cyanosis** (“**blue bloaters**”).
- ❖ **Long-standing severe** chronic bronchitis leads to
 - **cor pulmonale and cardiac failure.**
- ❖ Death may also result from
 - ❖ Impairment of respiratory function due to **superimposed acute infections.**

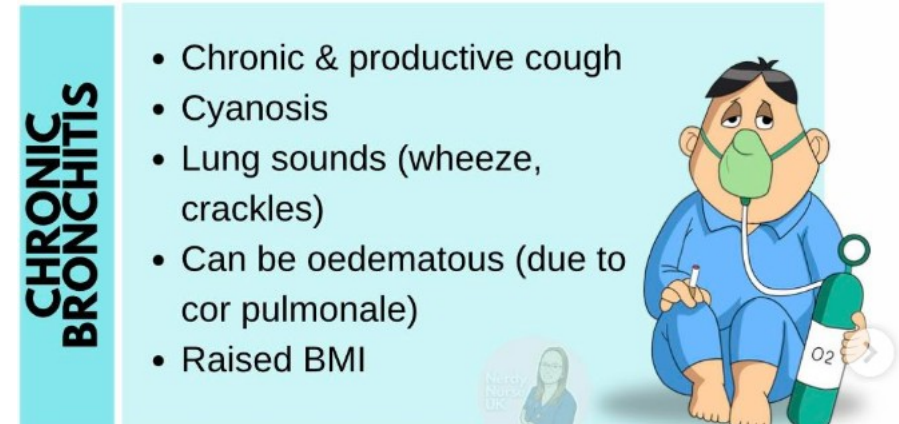


Table 15-4 Emphysema and Chronic Bronchitis

	Predominant Bronchitis	Predominant Emphysema
Age (yr)	40-45	50-75
Dyspnea	Mild; late	Severe; early
Cough	Early; copious sputum	Late; scanty sputum
Infections	Common	Occasional
Respiratory insufficiency	Repeated	Terminal
Cor pulmonale	Common	Rare; terminal
Airway resistance	Increased	Normal or slightly increased
Elastic recoil	Normal	Low
Chest radiograph	Prominent vessels; large heart	Hyperinflation; small heart
Appearance	Blue bloater	Pink puffer

Asthma

- ❖ Asthma is a **chronic disorder** of the **conducting airways**
- ❖ Usually caused by an **immunological reaction**
- ❖ Is marked by **episodic bronchoconstriction** due to **increased airway sensitivity** to a variety of stimuli
- ❖ Inflammation of the bronchial **walls** and **increased mucus secretion**.
- ❖ The disease is manifested by
 - Recurrent episodes of wheezing
 - Breathlessness, chest tightness
 - Cough, particularly at **night** and/or in the **early morning**.

1. Depending on the **frequency & severity** of symptoms

Mild

Moderate

Severe

- Rarely, a state of **unremitting attacks**, called **acute severe asthma**

2. Inciting agent

Intrinsic (non atopic)

Extrinsic (atopic)

In either type, episodes of bronchospasm can have **diverse triggers**, such as

respiratory infections (especially viral infections),
exposure to irritants (e.g., smoke, fumes), cold air, stress, and exercise.

Asthma may also be **classified** according to the **agents** or **events** that trigger bronchoconstriction. Include

- ✓ Seasonal
- ✓ Exercise-induced
- ✓ Drug-induced (e.g., aspirin)
- ✓ Occupational asthma
- ✓ Asthmatic bronchitis in smokers.

- ❖ Classic example of **IgE-mediated (type I) hypersensitivity** reaction
- ❖ Usually begins in **childhood**
- ❖ Triggered by **env'tal pollutants**
- ❖ Positive **family history** is common
- ❖ There is evidence of **allergen sensitization** and **immune** activation
- ❖ Asthmatic attacks often **preceded** by **allergic rhinitis, urticaria** or **eczema**
- ❖ Skin test is **positive** to allergen injection

Pathogenesis

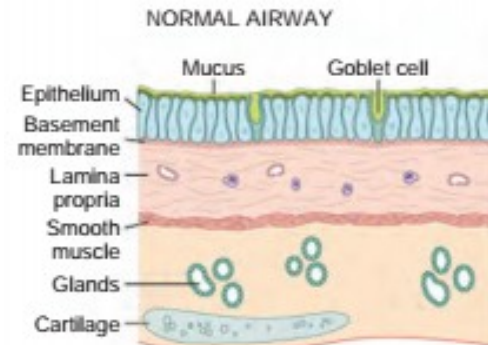
- ❖ The contributions of the **immune response**, **genetics** and **environment** are closely intertwined.
- ❖ **TH2 Responses, IgE and Inflammation.**
 - A fundamental abnormality in asthma is
 - An **exaggerated TH2** response to normally **harmless environmental antigens**
 - TH2 cells secrete **cytokines** that promote inflammation and stimulate **B cells** to produce **IgE** and other **antibodies**.
 - **IL-4**: stimulates the production of **IgE**
 - **IL-5**: activates locally recruited **eosinophils**
 - **IL-13**: stimulates **mucus secretion** and also **promotes IgE production by B cells**
- ❖ **IgE binds to the Fc receptors on submucosal mast cells**
 - **Repeat exposure** to the allergen triggers the mast cells to release granule contents and produce cytokines and other mediators
 - Induce the **early-phase (immediate hypersensitivity) reaction** and the **late phase reaction**.

Pathogenesis

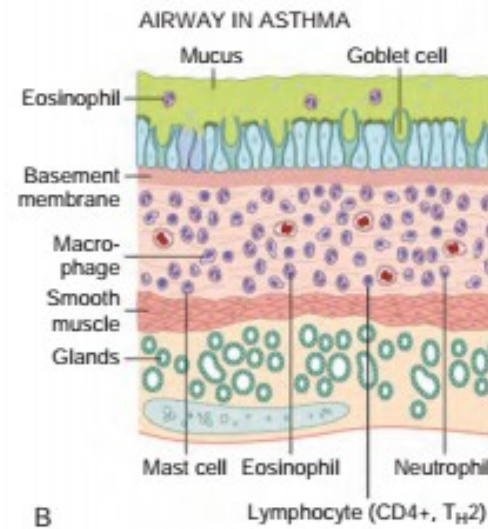
- ❖ The **early reaction** is dominated by
 - **Bronchoconstriction**
 - **Increased mucus production**
 - **Variable degrees of vasodilation**
 - **Increased vascular permeability**
- ❖ **Bronchoconstriction** is **triggered** by **direct stimulation of subepithelial vagal** (parasympathetic) receptors.
- ❖ The late-phase reaction is **dominated** by recruitment of **leukocytes**, notably **eosinophils, neutrophils**, and more **T cells**.

Mediators

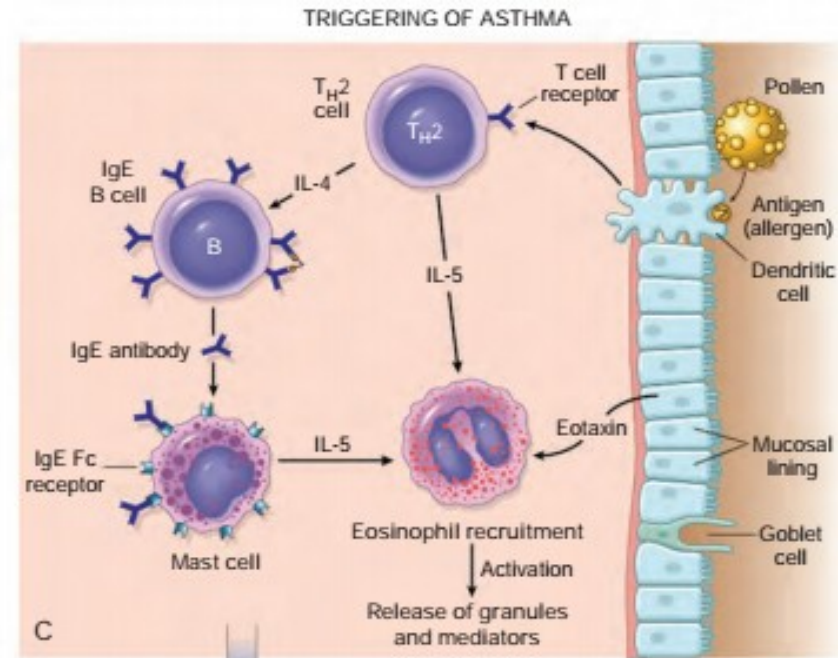
- ❖ **Leukotrienes C4, D4, and E4**, cause
 - **Prolonged bronchoconstriction**
 - **Increased vascular permeability**
 - **Increased mucus secretion**
- ❖ **Acetylcholine**: can cause airway **smooth muscle constriction** by directly **stimulating** muscarinic receptors
- ❖ **Histamine**: a potent bronchoconstrictor
- ❖ **Prostaglandin D2**: elicits bronchoconstriction and vasodilatation
- ❖ **Platelet-activating factor**: causes aggregation of platelets and release of serotonin from their granules



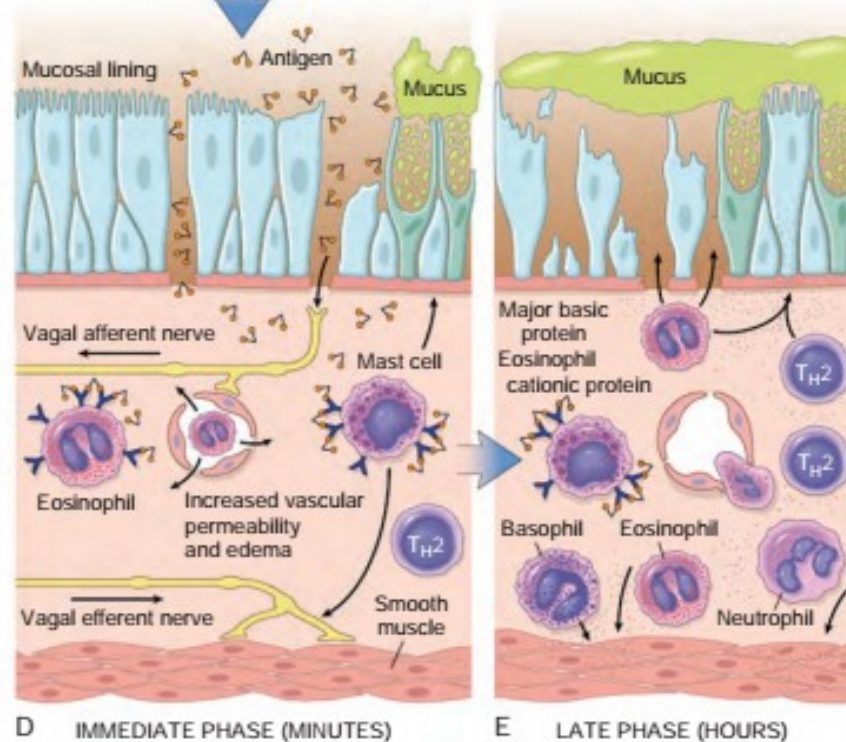
A



B



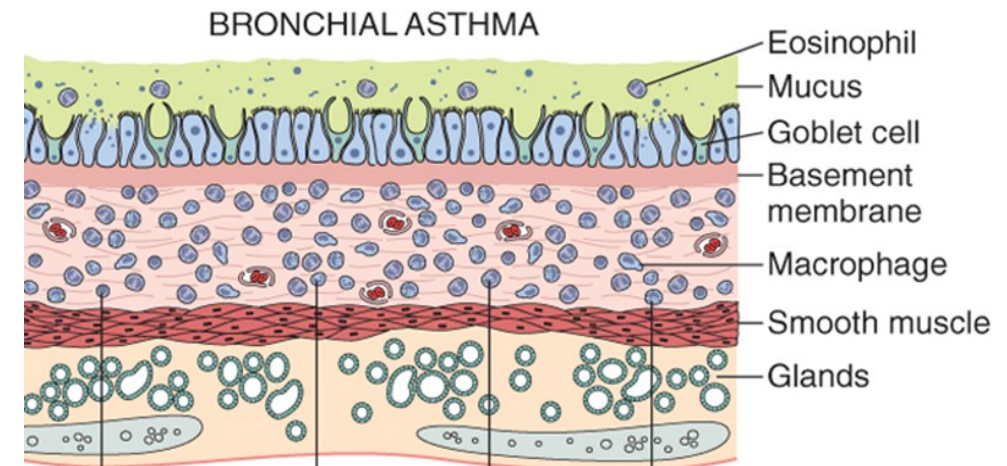
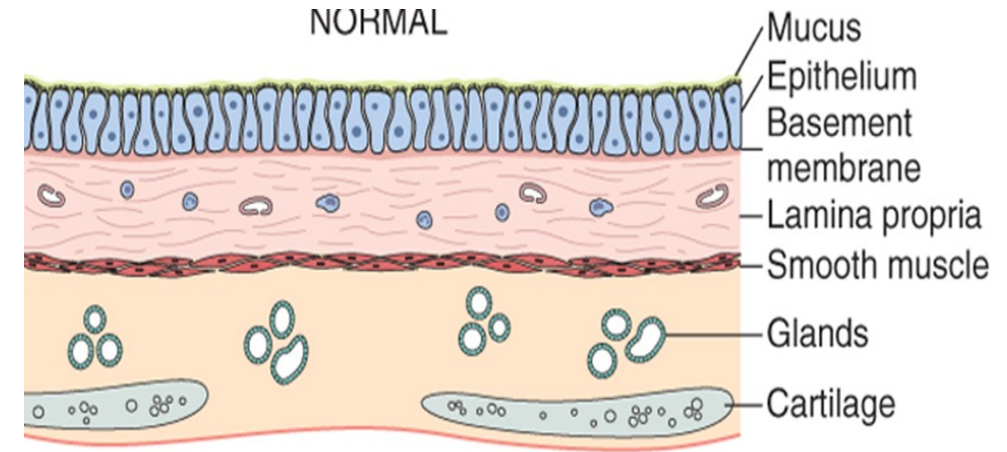
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MORPHOLOGY

- ❖ In patients dying of acute **severe asthma** (status asthmaticus) the lungs
 - Are **distended by overinflation**
 - contain small areas of **atelectasis**.
- ❖ The most striking gross finding is **occlusion** of bronchi and bronchioles by
 - **thick, tenacious mucus plugs**, which often contain **shed epithelium**.
- ❖ A characteristic finding in **sputum** or **bronchoalveolar lavage** specimens is **Curschmann spirals**
 - Result from **extrusion of mucus plugs** from subepithelial mucous gland ducts or bronchioles.
- ❖ Also present are numerous eosinophils
 - **Charcot-Leyden crystals**; are composed of an eosinophil protein called **galectin-10**.

- ❖ The other characteristic histologic findings (“**airway remodeling**”) include:
- Thickening of airway wall
 - Sub-basement membrane fibrosis (due to deposition of type I and III collagen)
 - Increased vascularity
 - An increase in the size of the **submucosal glands** and number of airway goblet cells
 - Hypertrophy and/or hyperplasia of the bronchial wall **muscle**



Clinical Course

- ❖ The cardinal symptoms
 - chest tightness
 - dyspnea
 - wheezing, and coughing (with or without sputum production)
- ❖ In its most **severe form, status asthmaticus**
 - airflow obstruction that is so extreme that **marked cyanosis or even death ensues.**
- ❖ The diagnosis is based on demonstration of an **increase in airflow obstruction**
 - Difficulty with exhalation (prolonged expiration, wheeze)
 - Peripheral blood eosinophilia
 - The finding of **eosinophils, Curschmann spirals, and Charcot-Leyden crystals** in the sputum
- ❖ Up to **50% of childhood asthma remits in adolescence.**

- ❖ **Do not** have evidence of allergen sensitization
- ❖ Most frequently triggered by **viral** respiratory infection
(e.g., rhinovirus, parainfluenza virus and respiratory syncytial virus)
- ❖ Positive family history is **uncommon**
- ❖ Serum **IgE** level is **normal**, **Skin test** is negative
- ❖ Virus induced inflammation of the respiratory mucosa lowers the threshold of the sub epithelial vagal receptors to irritants
- ❖ **Inhaled air pollutant** may also contribute

Drug induced Asthma

- ❖ Aspirin – tipping the balance toward **bronchoconstrictor leukotrienes**
- ❖ Aspirin (and other NSAIDs) triggers asthma in these patients by **inhibiting the cyclooxygenase leading to a rapid decrease in prostaglandin E2.**
 - Normally prostaglandin E2 inhibits enzymes that generate proinflammatory mediators such as leukotrienes B4, C4, D4 and E4

Occupational asthma- This form of asthma may be triggered by
Fumes (epoxy resins, plastics),
Organic and chemical dusts (wood, cotton, platinum),
Gases (toluene), or
Other chemicals (formaldehyde, penicillin products)

❖ Is a disorder in which destruction of **smooth muscle** and **elastic tissue** by chronic necrotizing infections leads to

➤ **permanent dilation** of **bronchi** and **bronchioles**

❖ Manifested by cough ,fever, & copious amounts of foul smelling position dependent purulent sputum

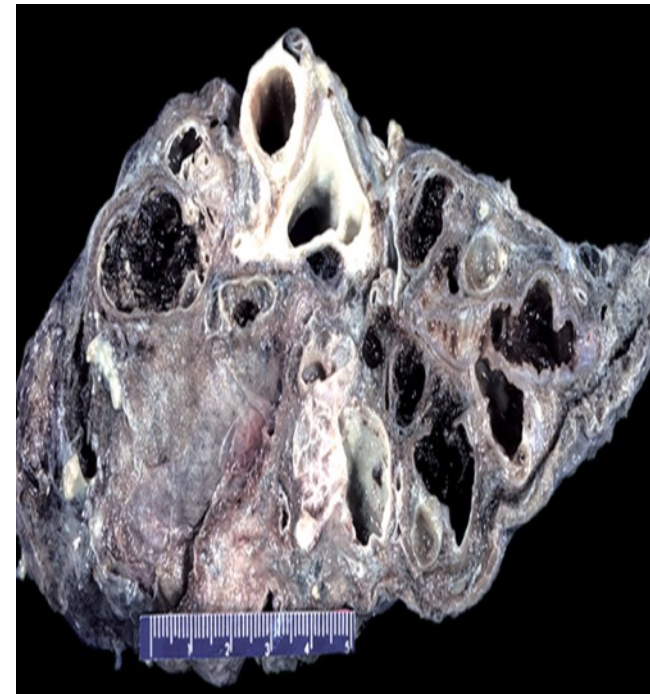
❖ Because of better control of lung infections, bronchiectasis is now uncommon.

Bronchiectasis

- ❖ It may develop in association with a variety of conditions, including
 - Cystic fibrosis
 - Intralobar sequestration of the lung
 - Immunodeficiency states
 - Primary ciliary dyskinesia
 - Kartagener syndrome
 - Infections, including necrotizing pneumonia caused by bacteria, viruses, or fungi
 - Bronchial obstruction, due to tumor, foreign bodies, or mucus impaction
 - Rheumatoid arthritis, SLE, IBD, COPD,
- ❖ One fourth to one half of cases are **idiopathic**.

MORPHOLOGY

- ❖ Usually affects the **lower lobes bilaterally**
- ❖ Is most severe in the more **distal bronchi and bronchioles**.
- ❖ The airways are dilated, sometimes up to four times normal size.
- ❖ On the cut surface of the lung, the dilated bronchi appear cystic and are filled with mucopurulent secretions
- ❖ The histologic findings vary with the activity and chronicity of the disease.



Clinical Course

- ❖ Bronchiectasis causes **severe, persistent cough; expectoration of foul smelling sputum**
- ❖ Obstructive respiratory insufficiency can lead to **marked dyspnea and cyanosis.**
- ❖ **Hemoptysis**
- ❖ **Systemic Symptoms:** Fatigue, weight loss, and night sweats.
- ❖ **Clubbing**

RESTRICTIVE LUNG DISEASES

Interstitial (Restrictive) Lung Diseases

- ❖ The primary physiological problem is **increased lung stiffness** (decreased compliance)
- ❖ **Restrictive lung disorders** occur in **two** general conditions:
 - **Chronic interstitial and infiltrative diseases**
 - Pneumoconioses and **interstitial fibrosis** of unknown etiology
 - **Chest wall disorders**
 - e.g., neuromuscular diseases such as poliomyelitis, severe obesity, pleural diseases, and kyphoscoliosis
- ❖ Chronic interstitial pulmonary diseases are a heterogeneous group of disorders
 - Characterized predominantly by **inflammation and fibrosis** of the **pulmonary interstitium**.
 - Many of the entities are of **unknown** cause
- ❖ Diffuse restrictive diseases are categorized based on **histology** and **clinical** features

Table 15-5 Major Categories of Chronic Interstitial Lung Disease

Fibrosing
Usual interstitial pneumonia (idiopathic pulmonary fibrosis) Nonspecific interstitial pneumonia Cryptogenic organizing pneumonia Connective tissue disease-associated Pneumoconiosis Drug reactions Radiation pneumonitis
Granulomatous
Sarcoidosis Hypersensitivity pneumonitis
Eosinophilic
Smoking Related
Desquamative interstitial pneumonia Respiratory bronchiolitis-associated interstitial lung disease
Other
Langerhans cell histiocytosis Pulmonary alveolar proteinosis Lymphoid interstitial pneumonia

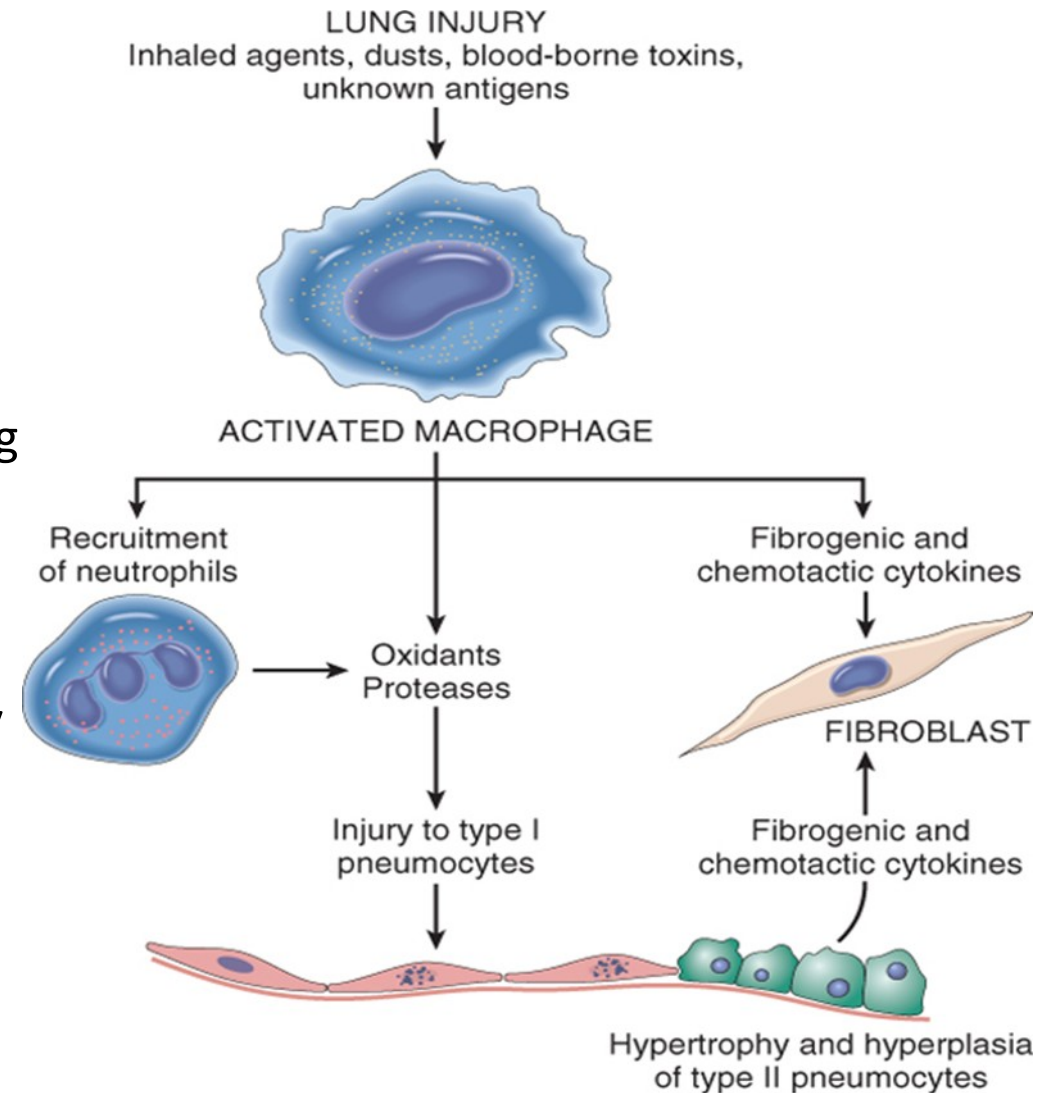
Fibrosing Diseases

Idiopathic Pulmonary Fibrosis

- ❖ IPF refers to a clinicopathologic syndrome marked by **progressive interstitial pulmonary fibrosis and respiratory failure**.
- ❖ In Europe the term *cryptogenic fibrosing alveolitis* is more popular.
- ❖ IPF has characteristic radiologic, pathologic, and clinical features.

Pathogenesis

- ❖ Arises in genetically predisposed individuals
- ❖ The **implicated factors** are :
 - *Environmental factors*
 - Most important among these is cigarette smoking
 - *Genetic factors*
 - *Age*
 - IPF is a disease of older individuals
- **Pathogenesis** of IPF is complex and poorly understood.



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MORPHOLOGY

❖ Grossly

- The pleural surfaces of the lung are **cobblestoned** as a result of the **retraction of scars** along the **interlobular septa**.

❖ Microscopically

- The hallmark is **patchy interstitial fibrosis (honeycomb fibrosis)**.
- There is **mild to moderate inflammation** within the fibrotic areas.
- Foci of **squamous metaplasia** and **smooth muscle hyperplasia** may be present,
- pulmonary arterial hypertensive changes (**intimal fibrosis** and **medial thickening**).

❖ In acute exacerbations **diffuse alveolar damage** may be **superimposed** on these chronic changes.

❖ **Pneumoconioses** (Occupational ILD): Due to inhalation of inorganic dusts.

- ❑ **Silicosis:** Nodular fibrosis from silica dust.

- ❑ **Asbestosis:** Diffuse interstitial fibrosis from asbestos fibers; associated with pleural plaques and lung cancer.

- ❑ Coal Worker's Pneumoconiosis (Black Lung Disease).

❖ In general, only a small percentage of exposed people develop occupational respiratory diseases.

Granulomatous Diseases

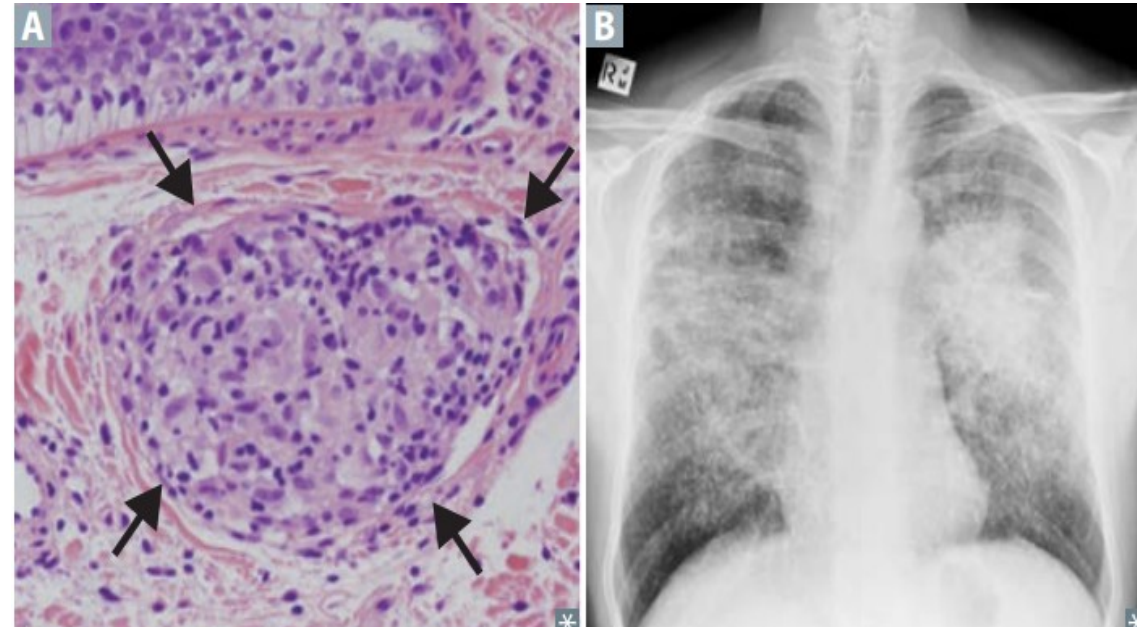
Sarcoidosis

- A multisystem **noncaseating granulomatous** disease of **unknown** etiology
- **Bilateral hilar lymphadenopathy** or lung involvement is most common(90%)
 - **visible** on chest radiographs
- Eye and skin involvement each occurs in about 25% of cases
 - may occasionally be the presenting feature of the disease.
- The diagnosis is one of exclusion

Sarcoidosis

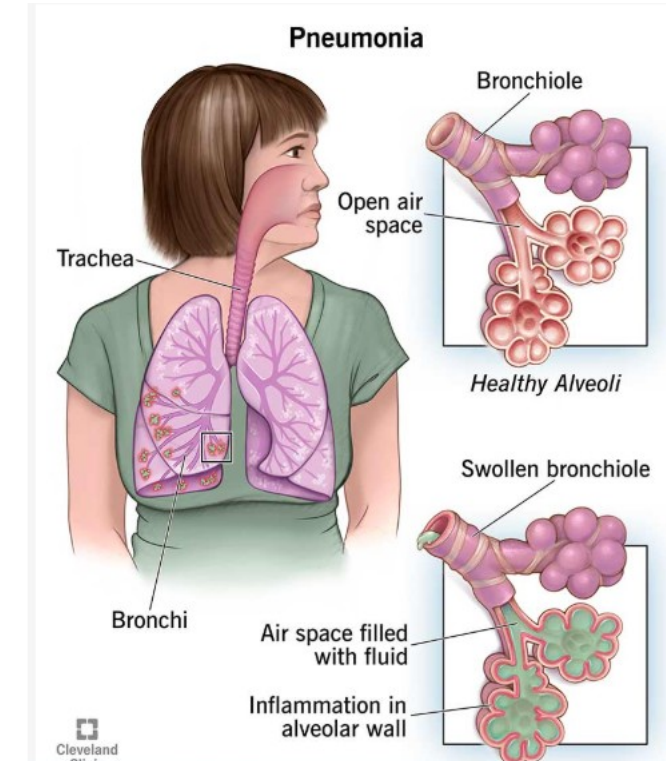
Morphology: regardless of the organ involved

- **Non caseating epithelioid granuloma**
- **Asteroid bodies**(inclusions in multinucleated giant cells)
- **Schaumann bodies: laminated concretions** composed of **calcium** and **proteins**
 - **not pathognomonic** of sarcoidosis
- The **lungs** are **common sites** of involvement
- **Lymph nodes** are involved in **almost all cases**
 - particularly the hilar and mediastinal nodes, but any LN



Pneumonia

- ❖ Inflammation of the lung parenchyma
- ❖ Pneumonia can result when
 - pulmonary **defense mechanisms** are impaired
 - when the **resistance of the host** in general is impaired
- e.g.*** Immunologic deficiency, immunosuppressive agents & leukopenia
- ❖ Pulmonary defense mechanisms (clearing mechanism) can be impaired
 - Loss or suppression of **cough reflex**
 - Coma, anesthesia, neuromuscular disorders, drugs, chest pain
 - Injury to **mucociliary apparatus**
 - Cigarette smoking, viral diseases, genetic disturbances
 - Interference of action of **alveolar macrophages**
 - Alcohol, tobacco, anoxia
 - Pulmonary **congestion & edema**
 - Accumulation of **secretions**
 - Bronchiectasis, cystic fibrosis



Classifications

- Pneumonias are classified on the basis of specific **etiologic agent** or **clinical setting**

Table 15-7 Pneumonia Syndromes

Community-Acquired Acute Pneumonia	Chronic Pneumonia
<i>Streptococcus pneumoniae</i> <i>Haemophilus influenzae</i> <i>Moraxella catarrhalis</i> <i>Staphylococcus aureus</i> <i>Legionella pneumophila</i> <i>Enterobacteriaceae</i> (<i>Klebsiella pneumoniae</i>) and <i>Pseudomonas</i> spp. <i>Mycoplasma pneumoniae</i> <i>Chlamydia</i> spp. (<i>C. pneumoniae</i> , <i>C. psittaci</i> , <i>C. trachomatis</i>) <i>Coxiella burnetii</i> (Q fever) Viruses: respiratory syncytial virus, parainfluenza virus and human metapneumovirus (children); influenza A and B (adults); adenovirus (military recruits);	<i>Nocardia</i> <i>Actinomyces</i> Granulomatous: <i>Mycobacterium tuberculosis</i> and atypical mycobacteria, <i>Histoplasma capsulatum</i> , <i>Coccidioides immitis</i> , <i>Blastomyces dermatitidis</i>
Health Care-Associated Pneumonia	Necrotizing Pneumonia and Lung Abscess
<i>Staphylococcus aureus</i> , methicillin-sensitive <i>Staphylococcus aureus</i> , methicillin-resistant <i>Pseudomonas aeruginosa</i> <i>Streptococcus pneumoniae</i>	Anaerobic bacteria (extremely common), with or without mixed aerobic infection <i>Staphylococcus aureus</i> , <i>Klebsiella pneumoniae</i> , <i>Streptococcus pyogenes</i> , and type 3 pneumococcus (uncommon)
Hospital-Acquired Pneumonia	Pneumonia in the Immunocompromised Host
Gram-negative rods, <i>Enterobacteriaceae</i> (<i>Klebsiella</i> spp., <i>Serratia marcescens</i> , <i>Escherichia coli</i>) and <i>Pseudomonas</i> spp. <i>Staphylococcus aureus</i> (usually methicillin-resistant)	Cytomegalovirus <i>Pneumocystis jiroveci</i> <i>Mycobacterium avium-intracellulare</i> Invasive aspergillosis Invasive candidiasis "Usual" bacterial, viral, and fungal organisms (listed herein)
Aspiration Pneumonia	
Anaerobic oral flora (<i>Bacteroides</i> , <i>Prevotella</i> , <i>Fusobacterium</i> , <i>Peptostreptococcus</i>), admixed with aerobic bacteria (<i>Streptococcus pneumoniae</i> , <i>Staphylococcus aureus</i> , <i>Haemophilus influenzae</i> , <i>Pseudomonas aeruginosa</i>)	

Community acquired acute pneumonia

- ❖ Acquired from the **normal environment**
 - bacterial or viral
- ❖ Bacterial infections often follows an **URTI viral** infection
- ❖ Bacterial invasion of the lung parenchyma causes
 - **alveoli** to be filled with an inflammatory **exudate**, causing **consolidation (solidification)** of the pulmonary tissue

Community acquired acute pneumonia

- ❖ Predisposing conditions include
 - extremes of **age**
 - **Chronic diseases** (congestive heart failure, COPD, and diabetes),
 - Congenital or acquired **immune deficiencies**
 - **Decreased** or **absent** splenic function

Causative bacteria

- **Streptococcus pneumoniae**
- **H. influenza**
- **S. aureus**
- **K. pneumoniae**
- **M. catarrhalis**
- **Pseudomonas spp**

Morphology

- ❖ Bacterial pneumonia has **two patterns** of **anatomic distribution**:
 - **Bronchopneumonia**
 - **Lobar pneumonia**
- ❖ **Patchy** consolidation of the lung is characteristic of **bronchopneumonia**
 - The **patchy** involvement may become confluent, producing **total lobar consolidation**
- ❖ **Lobar pneumonia** results in
 - Fibrinosuppurative consolidation of **large portion of a lobe** or of an **entire lobe**

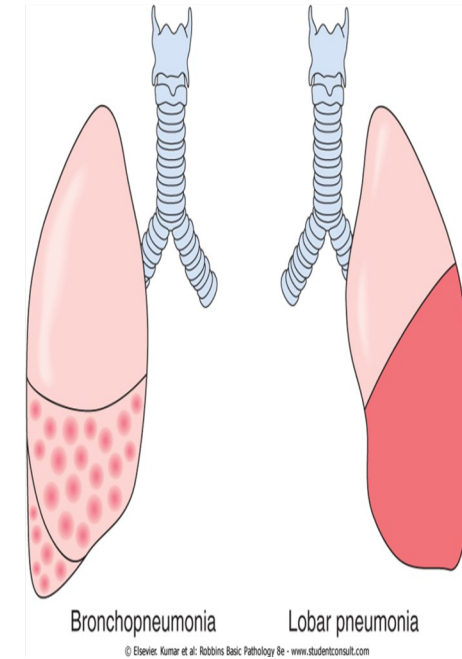


Figure 15-33 Bronchopneumonia. Section of lung showing patches of consolidation (arrows).



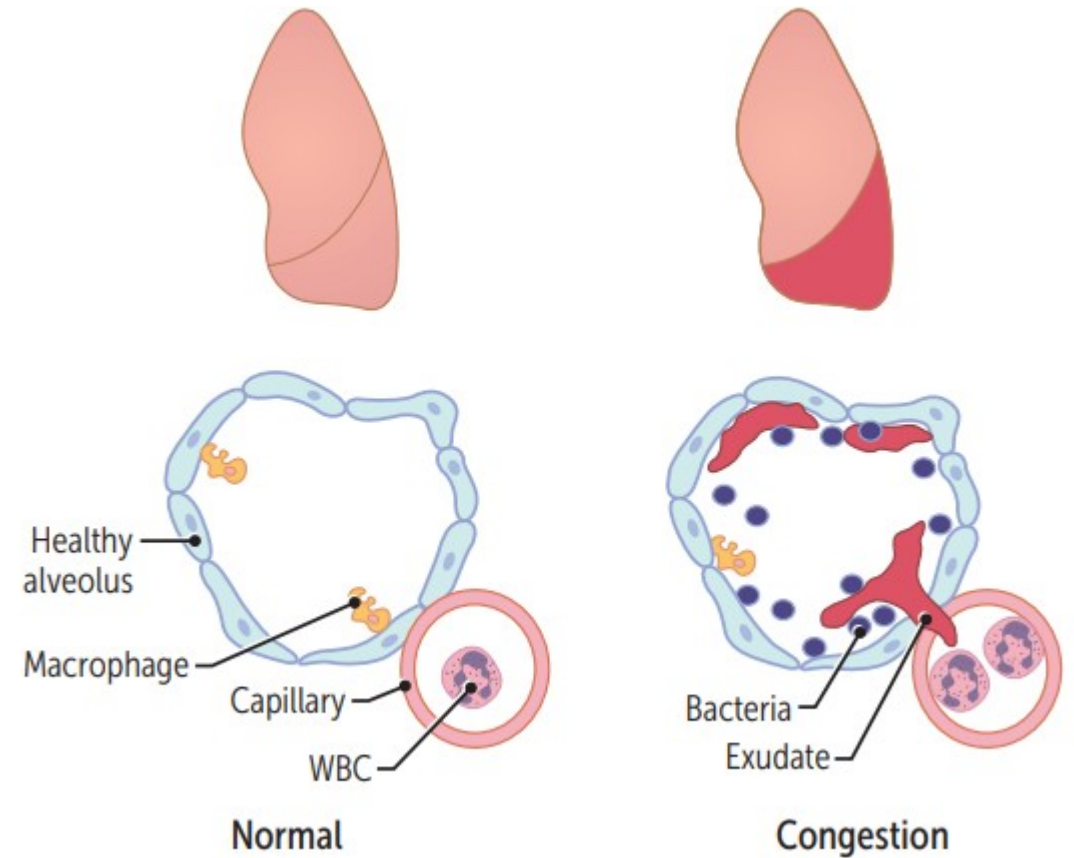
Figure 15-34 Lobar pneumonia—gray hepatization. The lower lobe is uniformly consolidated.

Ctd...

- ❖ One organism can produce either of the two
- ❖ These **anatomic categorizations** may be difficult to apply in individual cases because **patterns overlap**
- ❖ In lobar pneumonia, there are **four stages** of the inflammatory responses
 - Congestion
 - Red hepatization
 - Gray hepatization
 - Resolution

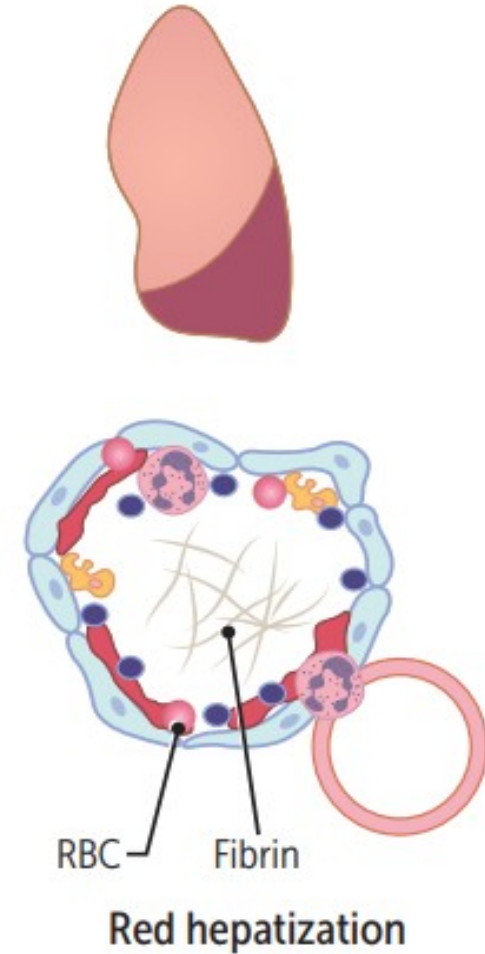
Congestion

- ❖ Lung is heavy, red & boggy
- ❖ Vascular engorgement
- ❖ Intra alveolar fluid with **few neutrophils**
- ❖ Presence of **numerous bacteria**



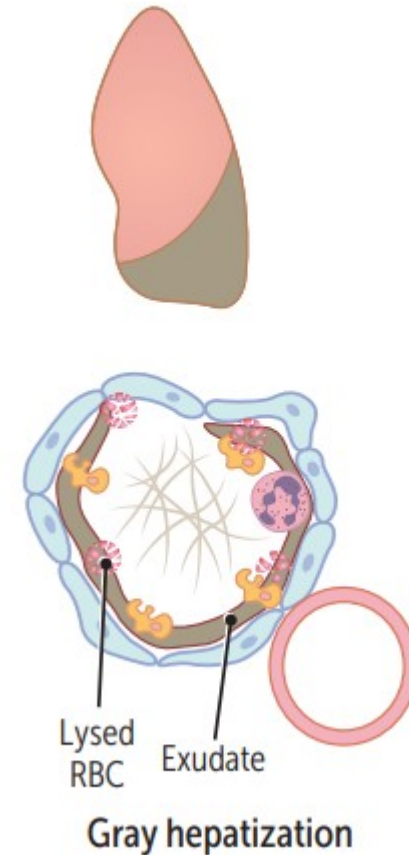
Red hepatization

- ❖ **Massive confluent exudation** with
 - red cells
 - neutrophils
 - fibrin filling alveolar spaces
- **Grossly**, the **lobe** appears as red, firm, & airless, with a **liver like consistency**



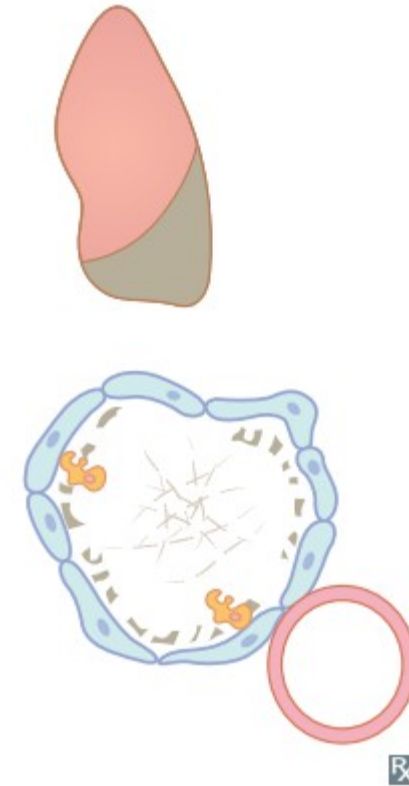
Gray hepatization

- ❖ Marked by
 - **Progressive disintegration of red cells**
 - The persistence of a fibrinosuppurative exudate
- ❖ **Results** in a color change to **grayish-brown**



Resolution

- ❖ Final stage => **resolution**
- ❖ The exudate **within the alveolar** spaces is broken **down by enzymatic digestion** to **produce** granular, semifluid debris that is
 - resorbed
 - ingested by macrophages
 - expectorated, or
 - organized by **fibroblasts** growing into it



Resolution

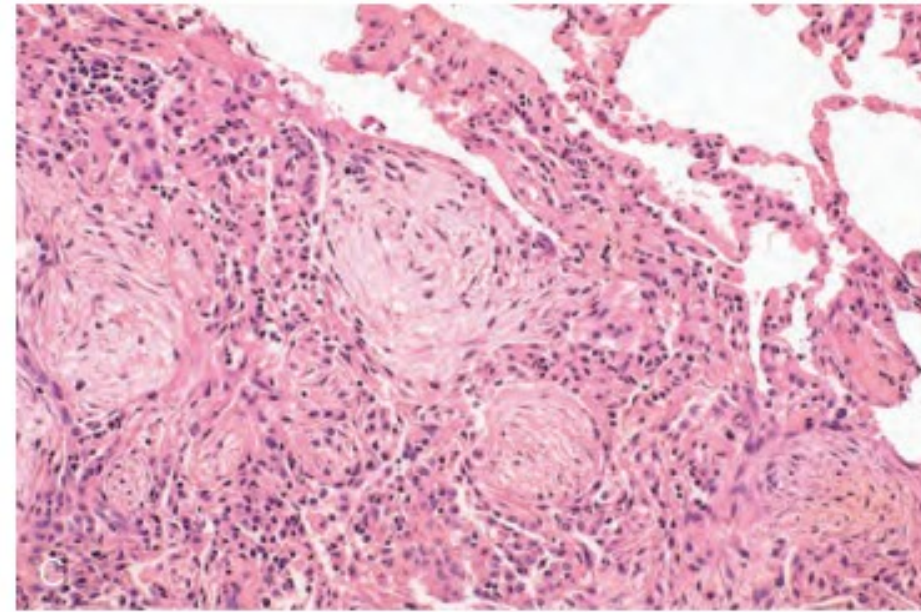
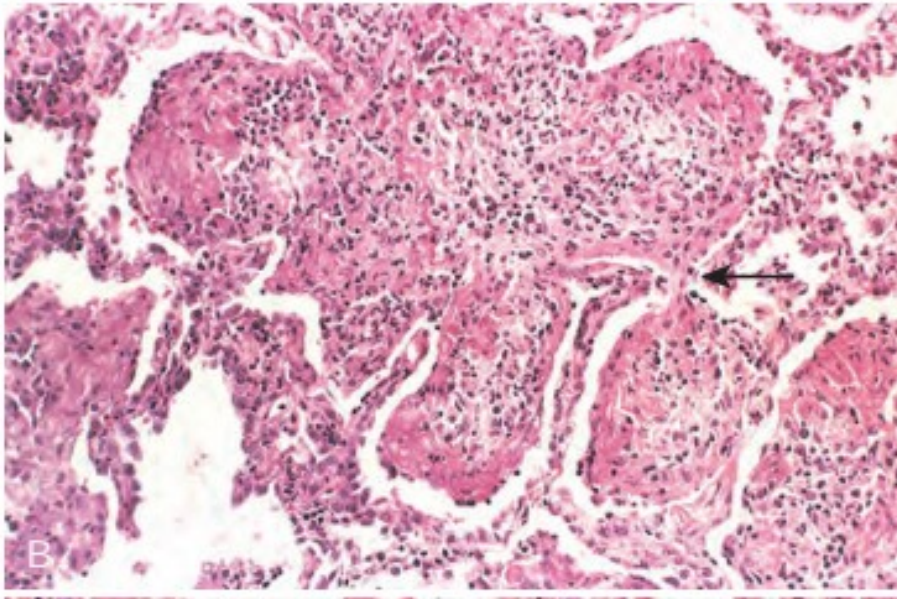
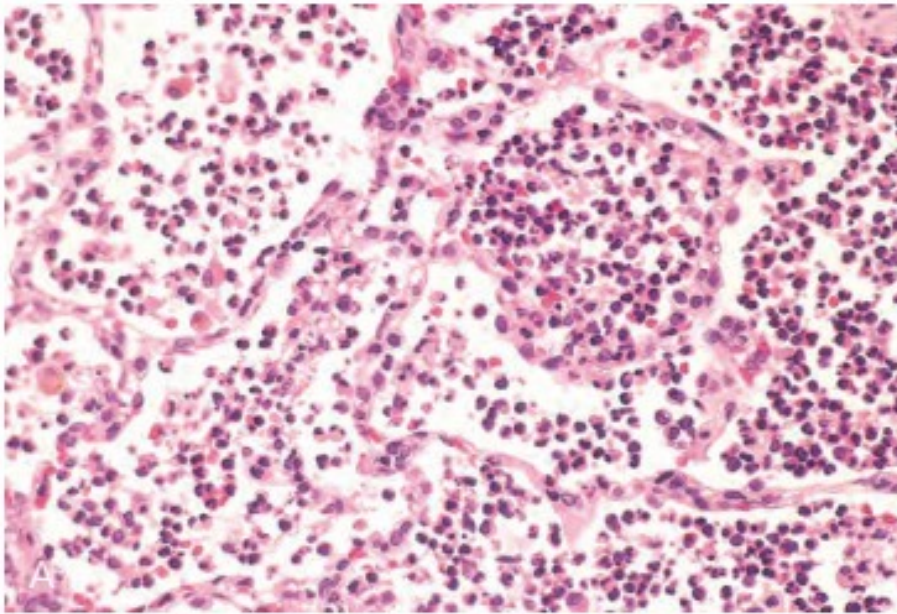


Figure 15-35 Stages of bacterial pneumonia. A, Acute pneumonia. The congested septal capillaries and numerous intra-alveolar neutrophils are characteristic of early red hepatization. Fibrin nets have not yet formed. **B,** Early organization of intra-alveolar exudate, seen focally to be streaming through the pores of Kohn (*arrow*). **C,** Advanced organizing pneumonia. The exudates have been converted to fibromyxoid masses rich in macrophages and fibroblasts.

Complications

❖ Complications of pneumonia include

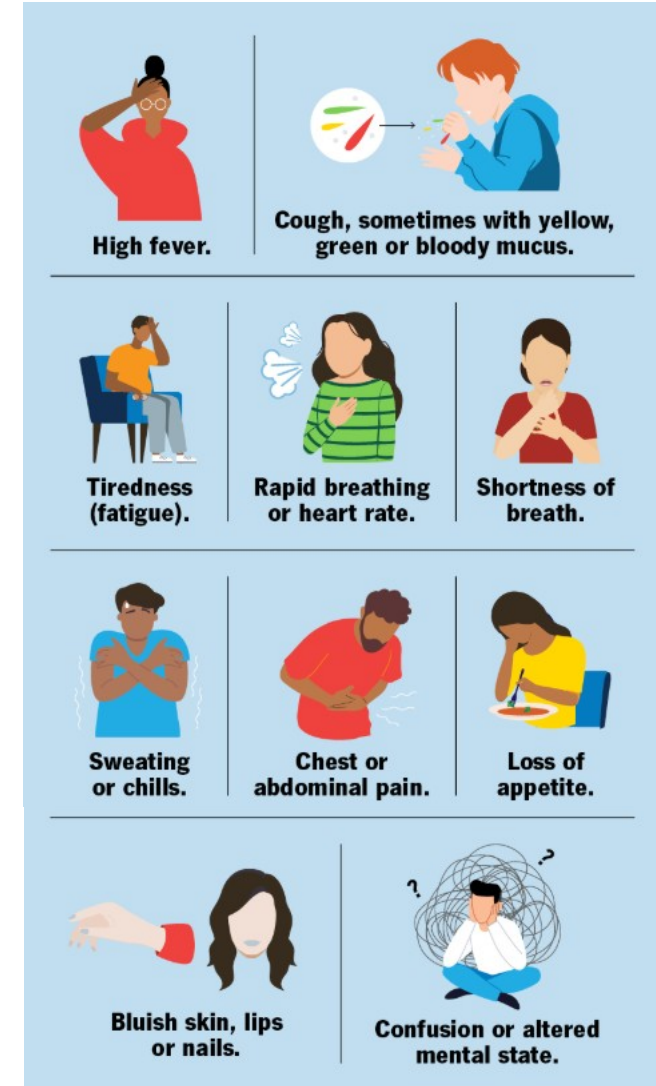
- Tissue destruction and necrosis, causing **abscess formation**
 - Particularly common with **type 3 pneumococci** or *Klebsiella* infections
- **Spread** of infection to the **pleural cavity**, causing the intrapleural fibrinosuppurative reaction known as **empyema**
- **Bacteremic dissemination** to
 - heart valves, pericardium
 - brain, kidneys, spleen, or joints

=> **metastatic abscesses, endocarditis, meningitis, or suppurative arthritis.**

Clinical Course of typical pneumonia

- ❖ Sudden onset of high-grade fever and chills.
- ❖ **Productive cough** with purulent (yellow/green) or "rust-colored" sputum (classic for pneumococcal pneumonia).
- ❖ Pleuritic chest pain (sharp, worse on inspiration).
- ❖ Dyspnea.

Signs: Tachypnea, tachycardia, dullness to percussion, **bronchial breath sounds, egophony** (E to A change), **crackles/rales**.



Atypical pneumonia

- ❖ Characterized by **patchy** inflammatory changes in the lungs
 - largely confined to the **alveolar septa** & pulmonary **interstitium**
 - Alveolar septa are widened & edematous, infiltrated by lymphocytes, histiocytes, plasma cells
- ❖ Atypical – denotes
 - Prominent extrapulmonary symptoms: headache, myalgias, fatigue, sore throat.
 - Dry, non-productive cough.
 - No physical findings of consolidation
 - Only moderate elevation of white cell count
 - Lack of alveolar exudate
- ❖ Causes are **mycoplasma, viruses, chlamydia**

Nosocomial pneumonia

- ❖ Defined as pulmonary infections acquired in the **course of a hospital stay**
- ❖ Common in pts with
 - severe underlying disease
 - prolonged antibiotic therapy
 - immunosuppression
 - invasive access devices such as intravascular catheters
- ❖ Patients on mechanical ventilation are at particularly high risk.
- ❖ **S. aureus, enterobactreciae, pseudomonas** are the commonly isolated organisms
- ❖ Nosocomial infections are **serious & life threatening** infections

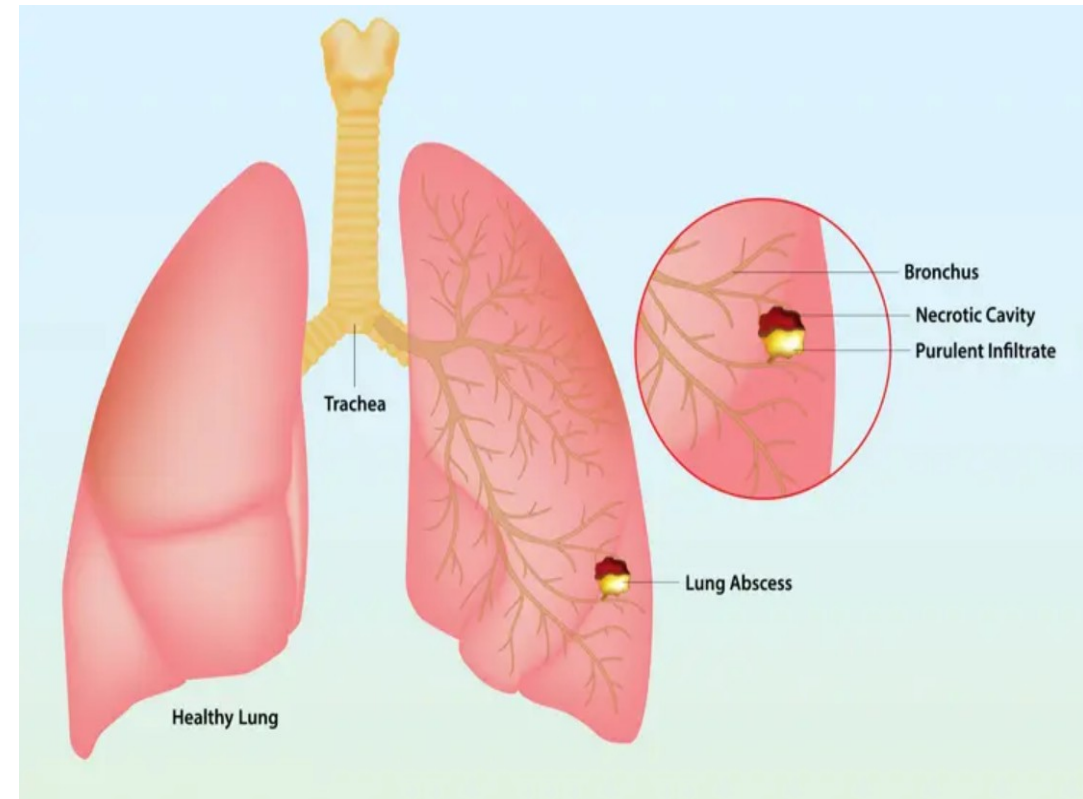
Aspiration pneumonia

- ❖ Occurs in **markedly debilitated** pts or who **aspirate gastric contents** while
 - Unconscious (after stroke)
 - During repeated vomiting
- ❖ It is a result of **chemical inflammation & infection with normal oral flora**
- ❖ Often necrotizing, pursues a fulminant clinical course
- ❖ Frequent cause of death
- ❖ **Micro-aspiration**, in contrast, occurs frequently in almost all people
 - especially those with **GERD**
- In those who survive, **lung abscess** formation is a common complication

Lung abscess

- ❖ A local suppurative process within the lung, characterized by
 - **Necrosis** of the pulmonary tissue
 - Formation of **cavities more than 2cm** containing necrotic debris
- ❖ This is often caused by **aspiration**
- ❖ **Alcoholism** is the most common condition predisposing to lung abscess
- ❖ Lung abscess is considered
 - **Primary** when it resulted from **existing lung parenchymal process**
 - **Secondary** when it complicates another process

e.g., Vascular emboli or follows **rupture of extrapulmonary abscess** into the lung



Etiologies

- ❖ Are **mixed** flora of aerobic & anaerobic
- ❖ Aerobic – S. aureus, klebsiella, haemophilus, pseudomonas, nocardia, E. coli, streptococcus, mycobacteria
- ❖ Oral cavity anaerobes – bacteroids, **fusobacterium & peptococcus spp**
- ❖ Fungi – candida, aspergillus
- ❖ Parasites – E. histolytica

Morphology

- ❖ Abscesses vary in diameter from a few millimeters to large cavities of 5 to 6 cm
- ❖ Due to aspiration, more common on the **right** and **single**
- ❖ Abscess in the case of **bronchiectasis** & **pneumonia** are usually **multiple & basal**
- ❖ **Septic emboli** and **pyemic abscesses** are **multiple** and may affect any region of the lungs.
- ❖ The cardinal histologic change in all abscess is
 - Suppurative destruction of lung parenchyma within the central area of cavitations

Lung abscess

Clinical course

- ❖ Clubbing of the fingers & toes
- ❖ It is always important to exclude **cancer** whenever there is lung abscess
- ❖ Can be Complicated by
 - Septic emboli
 - infection of pleural cavity
 - brain abscess
 - rarely amyloidosis



Chronic pneumonia

- ❖ Localized lesion in the **immunocompetent** pts
 - with or without regional lymph node involvement
- ❖ Typically, the inflammatory reaction is **granulomatous** , and is caused by
 - Bacteria (e.g., *M. tuberculosis*)
 - Fungi (e.g., *Histoplasma capsulatum*).

Pneumonia in the immunocompromised host

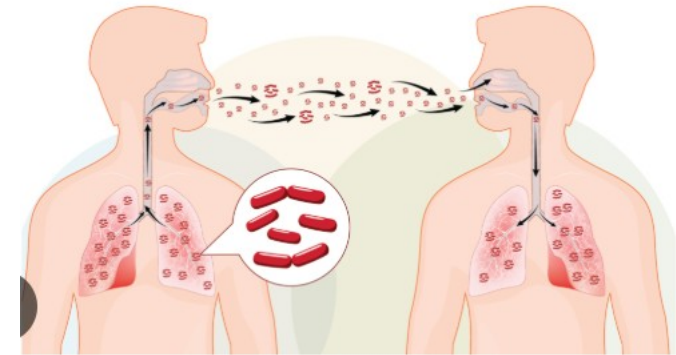
- ❖ Classified according to the etiologic agent:
 - Bacteria – p.aeruginosa, mycobacteria species
 - Viruses – CMV, herpes virus
 - Fungi – P. carinii, candida, aspergillus

TUBERCULOSIS

- ❖ A serious **chronic communicable granulomatous pulmonary** and **systemic** disease caused most often by *M. tuberculosis*.
- ❖ Usually involves the lungs but may affect **any organ or tissue** in the body.
- ❖ The source of transmission is
 - Humans with **active** tuberculosis who release mycobacteria present in **sputum**
 - By drinking **milk** contaminated with *M. bovis*
- ❖ It is important that **infection** with *M. tuberculosis* be differentiated from **active** disease.
- ❖ Infection typically leads to the development of **delayed hypersensitivity** to *M. tuberculosis* antigens, which can be detected by the **tuberculin (PPD, or Mantoux) skin test**.

Transmission

- ❖ Infections are acquired by direct person-to-person transmission of **airborne droplets**
 - Generated by **expectoration**
- ❖ **Oropharyngeal** and **intestinal** tuberculosis is contracted by drinking **milk** contaminated with ***Mycobacterium bovis***



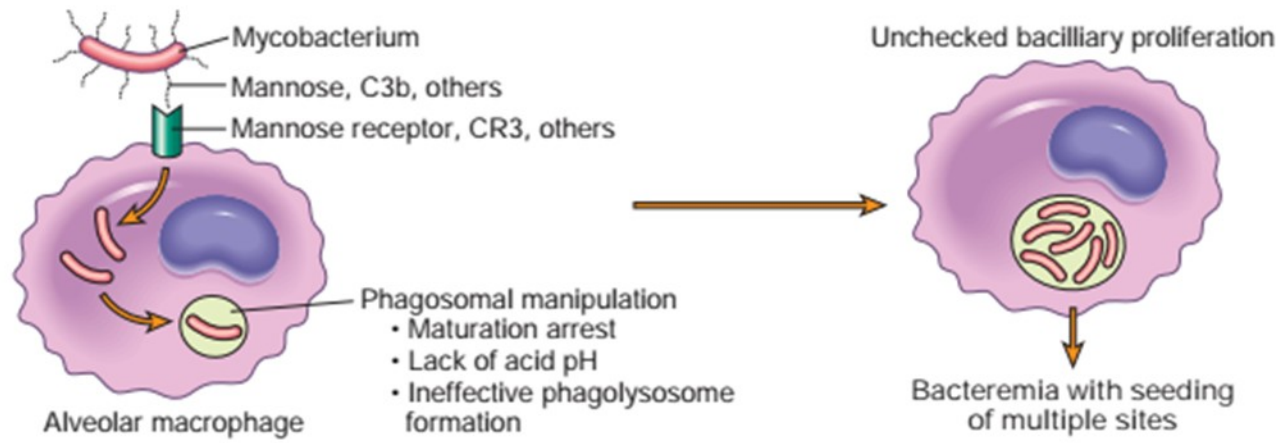
Etiology

- ❖ Mycobacteria are **slender rods** that are **acid fast**
- ❖ *M. tuberculosis hominis* is responsible for most cases of tuberculosis
- ❖ *Mycobacterium bovis* – **oropharyngeal & intestinal** tuberculosis
- ❖ Both species are **obligate aerobes** whose slow growth is retarded by a pH lower than **6.5**.
- ❖ *M. avium-intracellulare*, are much less virulent than *M. tuberculosis*

Pathogenesis

- ❖ The outcome of infection in a previously unexposed, immunocompetent person depends on the development of **anti-mycobacterial T-cell-mediated** immunity.
- ❖ Infection by *M. tuberculosis* proceeds in steps
 - From initial infection of macrophages => to a subsequent **TH1 response** that both **contains the bacteria** and **causes tissue damage**
- ❖ Early in infection, *M. tuberculosis* replicates **essentially unchecked** within macrophages
- ❖ Later in infection, the **cell response** stimulates macrophages to contain the proliferation of the bacteria

A. INFECTION BEFORE ACTIVATION OF CELL MEDIATED IMMUNITY



B. INITIATION AND CONSEQUENCES OF CELL MEDIATED IMMUNITY

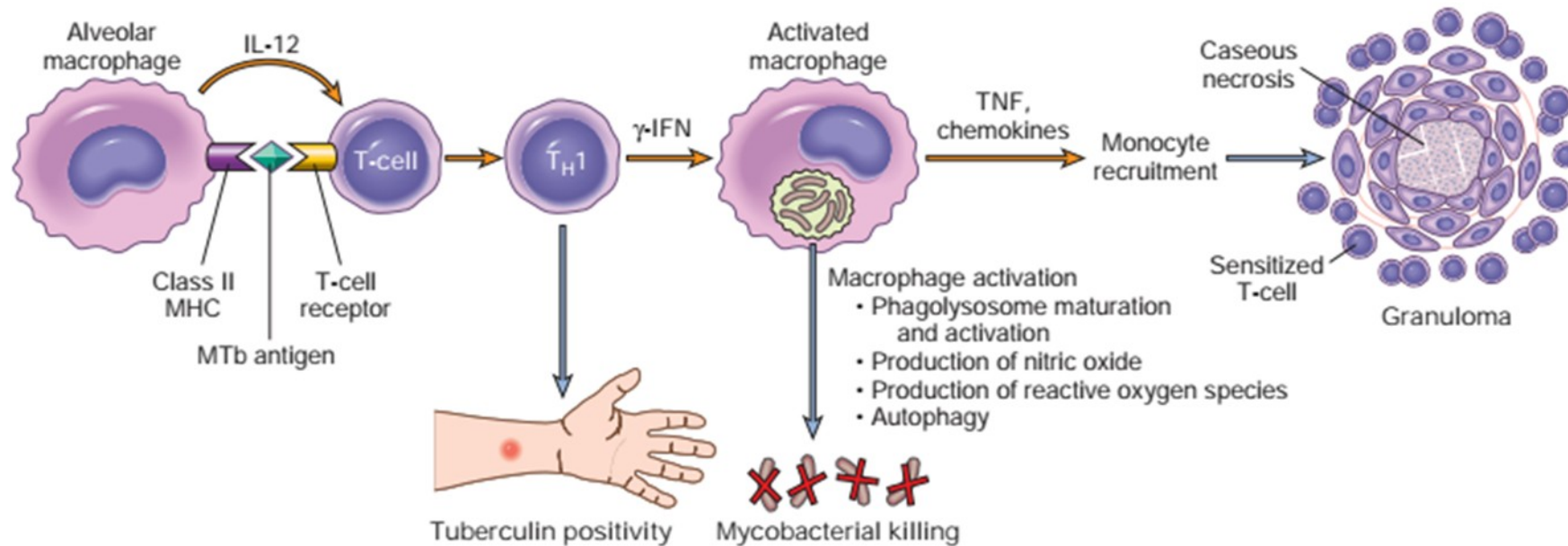


Figure 8-24 The sequence of events in primary pulmonary tuberculosis, commencing with inhalation of virulent *Mycobacterium tuberculosis* organisms and culminating with the development of cell-mediated immunity to the organism. **A**, Events occurring during early infection, before activation of T-cell-mediated immunity. **B**, The initiation and consequences of T-cell-mediated immunity. The development of resistance to the organism is accompanied by the appearance of a positive tuberculin test. γ -IFN, interferon- γ ; MHC, major histocompatibility complex; MTB, *M. tuberculosis*; TNF, tumor necrosis factor.

- ❖ In many people this response **halts** the infection before significant tissue destruction or illness occur.
- ❖ In other people the infection progresses due to
 - **Advanced age**
 - **Immunosuppression**
- ❖ The ongoing immune response results in **caseating necrosis**.
- ❖ Activated macrophages also secrete **TNF** and **chemokines**, which promote recruitment of more **monocytes**.
- ❖ Host susceptibility to disease
 - People with genetic **deficiencies** in the **IL-12 pathway** and the **IFN- γ pathway**, are **vulnerable** to severe mycobacterial infections.
- ❖ Immunological state in active tuberculosis
 - It is not entirely clear why some people progress from latent to active tuberculosis.

Ctd...

- ❖ The pathologic features of tuberculosis, such as
 - **Caseating granulomas** and **cavitation**, are the result of the **destructive tissue hypersensitivity** that is part of the **host immune response**.

Clinical Features

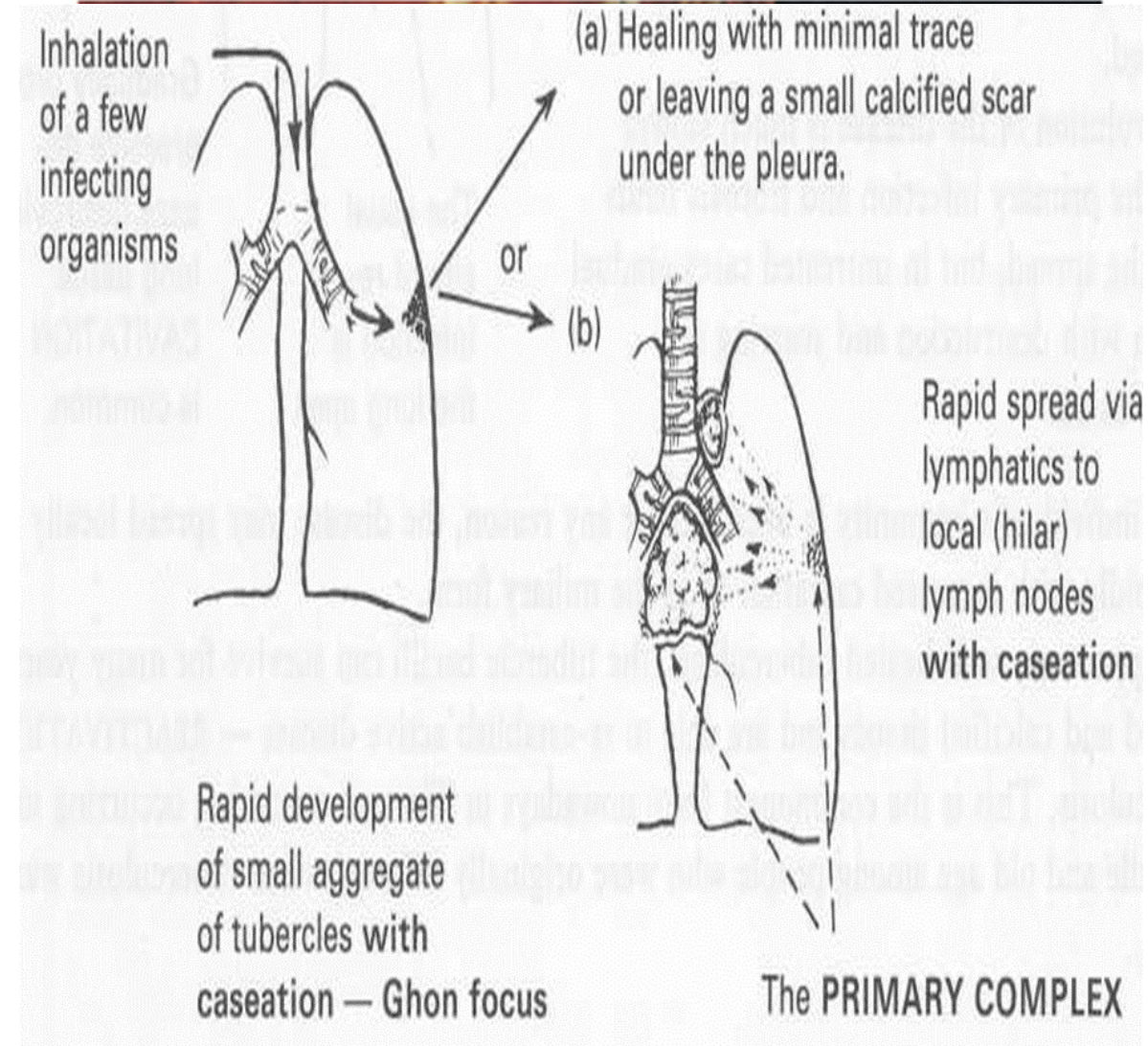
- ❖ Clinical tuberculosis is separated into **two** important pathophysiologic types:
 - “Primary” tuberculosis, which occurs in the **non-immune host**
 - “Secondary” tuberculosis, which occurs in the host who is **immune** to *M. tuberculosis*.

Primary Tuberculosis

- ❖ The form of disease that develops in a **previously unexposed/unsensitized**, person.
- ❖ The source of the organism is **exogenous**.
- ❖ About **5%** of those **newly infected** develop significant disease.
- ❖ In most people, the primary infection is **contained**, but in others, primary tuberculosis is progressive.
- ❖ The diagnosis of progressive primary tuberculosis in adults can be difficult.
- ❖ In contrast to secondary tuberculosis (**apical disease** with **cavitation**)
 - Progressive primary tuberculosis more often resembles an ***acute bacterial pneumonia*** with consolidation of the lobe, hilar adenopathy, and pleural effusion.
- ❖ **Lymphohematogenous dissemination** following primary infection may result in the development of **tuberculous meningitis** and **miliary tuberculosis**

Morphology

- ❖ Primary tuberculosis almost always begins in the **lungs**.
- ❖ The inhaled bacilli implant in the **distal** airspaces of the **lower part of the upper lobe** or the **upper part of the lower lobe**, usually **close to the pleura**.
- ❖ As sensitization develops, a **1- to 1.5-cm** area of gray-white inflammatory consolidation emerges => The **Ghon focus**



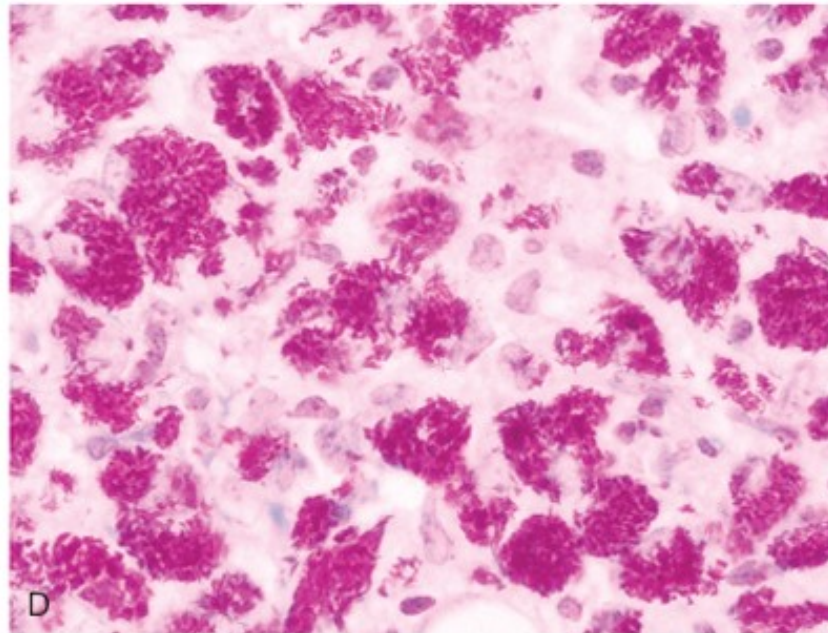
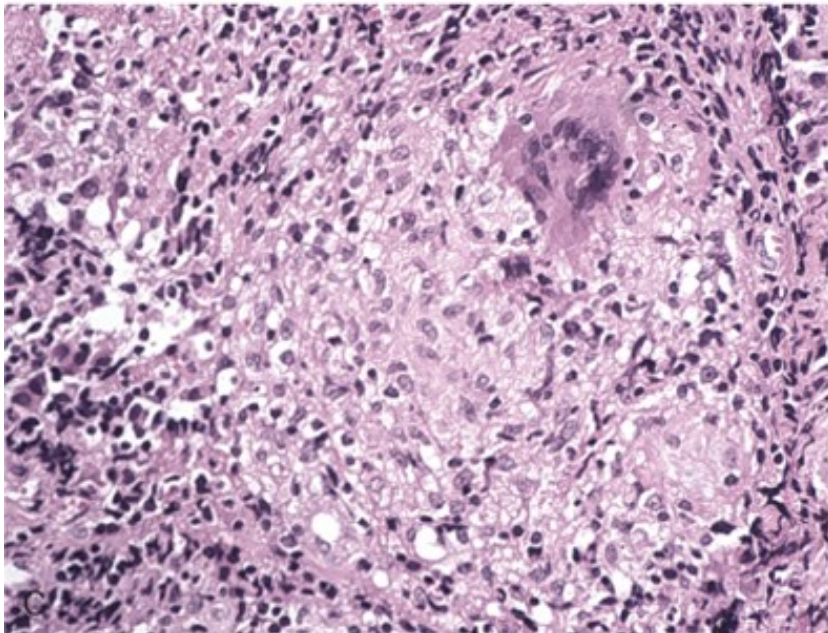
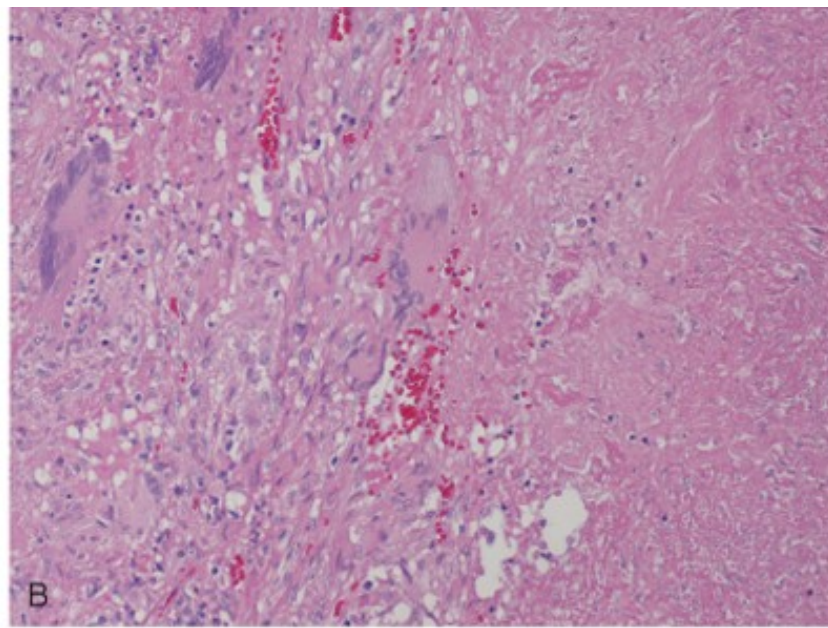
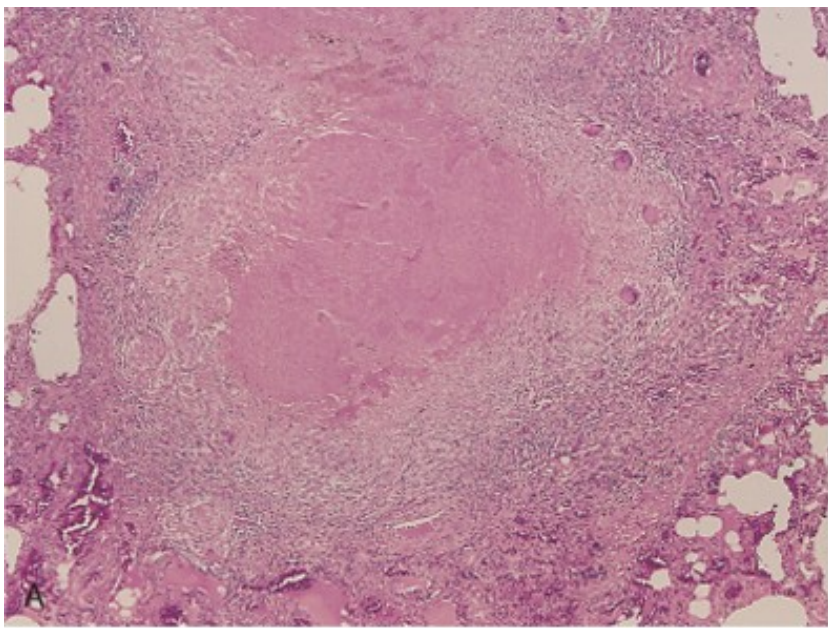
parenchymal focus is under the pleura in the lower part of the upper lobe (red arrow). Hilar lymph nodes with caseation are seen on the left (blue arrow).

TUBERCULOSIS

- ❖ Tubercle bacilli, either free or within phagocytes, drain to the **regional nodes**, which also often **caseate**.
- ❖ This combination of **parenchymal** lung lesion and **nodal** involvement is referred to as the **Ghon complex**.
- ❖ During the first few weeks there is also **lymphatic** and **hematogenous dissemination** to other parts of the body.
- ❖ In approximately **95%** of cases, development of **cell-mediated immunity controls** the infection.
 - **Ghon complex** undergoes **progressive fibrosis**, often followed by radiologically detectable **calcification (Ranke complex)**.

Histology

- ❖ Sites of active involvement are marked by
 - A characteristic **granulomatous** inflammatory reaction that forms both **caseating** and **noncaseating** tubercles .
- ❖ The granulomas are usually enclosed within a **fibroblastic rim** punctuated by **lymphocytes**.
- ❖ **Multinucleated giant cells** are present in the granulomas.
- ❖ **Immunocompromised** people **do not** form the characteristic granulomas and their macrophages contain many bacilli.



Secondary Tuberculosis (Reactivation Tuberculosis)

- ❖ Is the pattern of disease that arises in a **previously sensitized host**.
- ❖ It may also result from **exogenous re-infection**
 - **waning** of the protection afforded by the primary disease
 - a **large inoculum** of virulent bacilli.
- ❖ Classically localized to the **apex** of one or both upper lobes.
- ❖ **Cavitation** occurs readily in the **secondary** form
- ❖ Should always be an important consideration in **HIV-positive** patients who present with pulmonary disease.

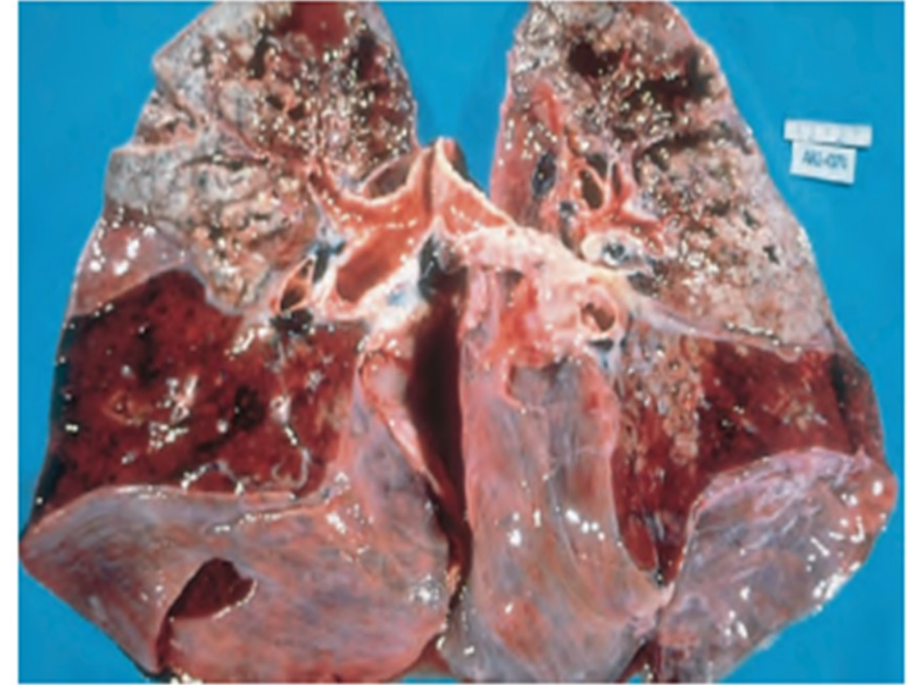
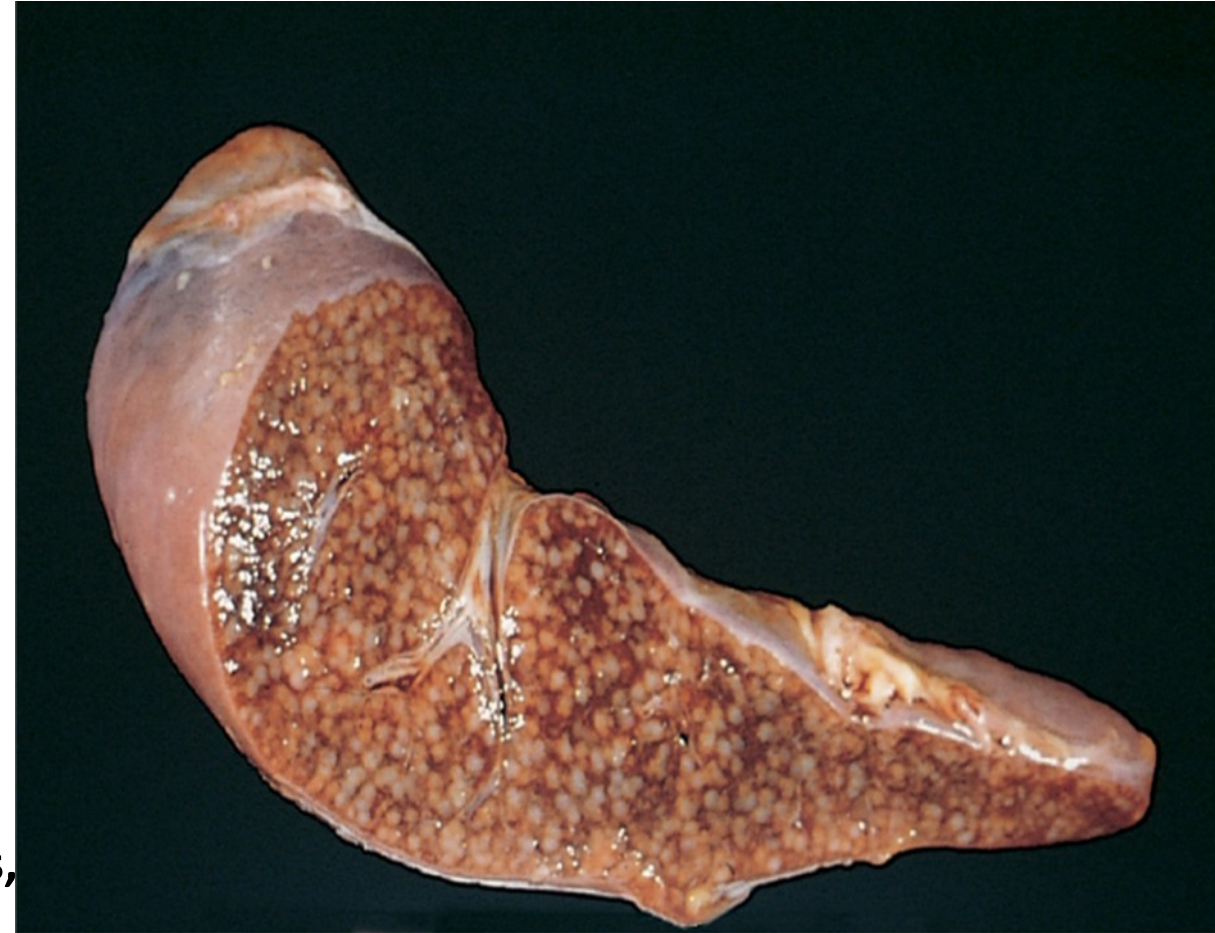


Figure 8-28 Secondary pulmonary tuberculosis. The upper parts of both lungs are riddled with gray-white areas of caseation and multiple areas of softening and cavitation.

Miliary pulmonary disease

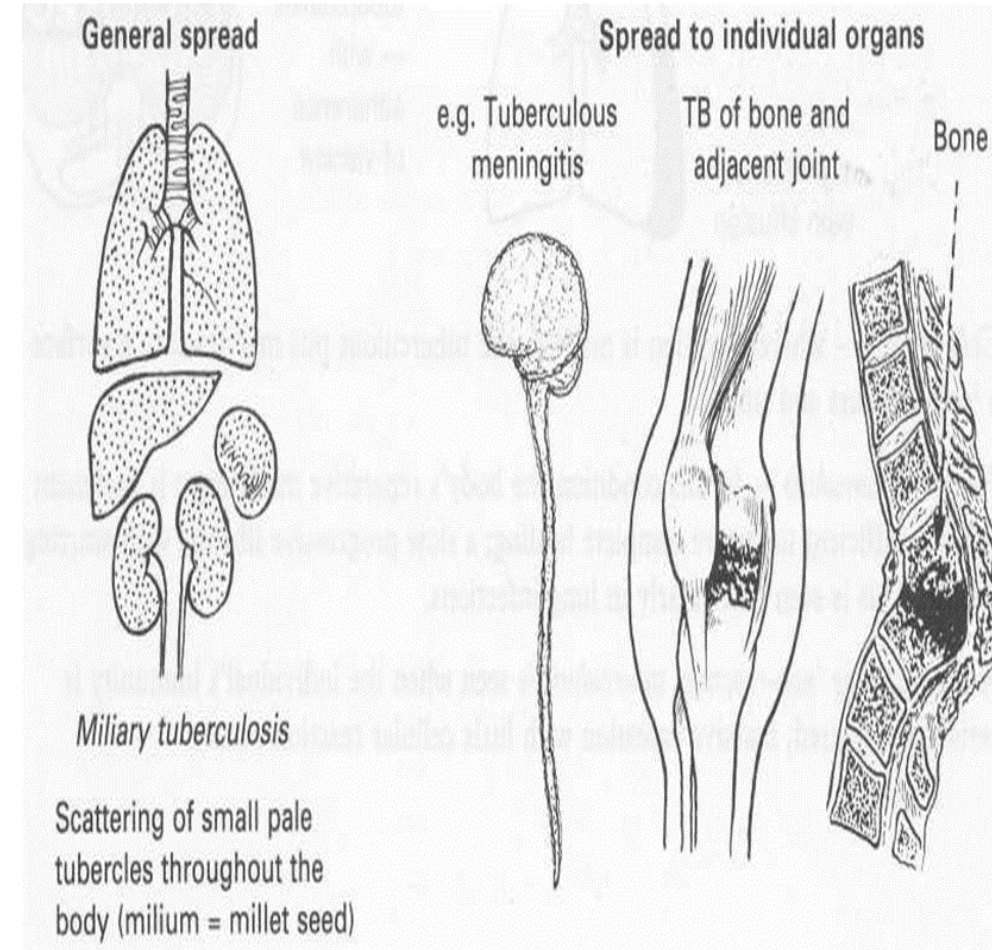
- ❖ Occurs when organisms draining through lymphatics enter the **venous blood** and **circulate back to the lung**.
- ❖ Individual lesions are either **microscopic** or **small, visible (2-mm)** foci of **yellow white** consolidation
- ❖ **Systemic** miliary tuberculosis occurs when bacteria disseminate through the **systemic arterial system**.
- ❖ Most prominent in the
 - **liver, bone marrow, spleen, adrenals, meninges, kidneys, fallopian tubes, and epididymis**



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Isolated tuberculosis

- ❖ Organs that are commonly involved include
 - Meninges (tuberculous meningitis)
 - Kidneys (renal tuberculosis)
 - Adrenals
 - Bones (osteomyelitis)
 - Fallopian tubes (salpingitis)
- ❖ When the **vertebrae** are affected, the disease is referred to as **Pott disease**.
- ❖ **Paraspinal “cold” abscesses** - may track along tissue planes and present as an **abdominal** or **pelvic mass**.



TUBERCULOSIS

- ❖ **Lymphadenitis** is the **most frequent** presentation of **extrapulmonary** tuberculosis, usually occurring in the **cervical** region (“**scrofula**”).
- ❖ In **HIV-negative** individuals, lymphadenitis tends to be **unifocal and localized**.
- ❖ HIV-positive people
 - almost always have **multifocal** disease
 - systemic symptoms
 - either pulmonary or other organ involvement by active tuberculosis



TUBERCULOSIS

Intestinal tuberculosis

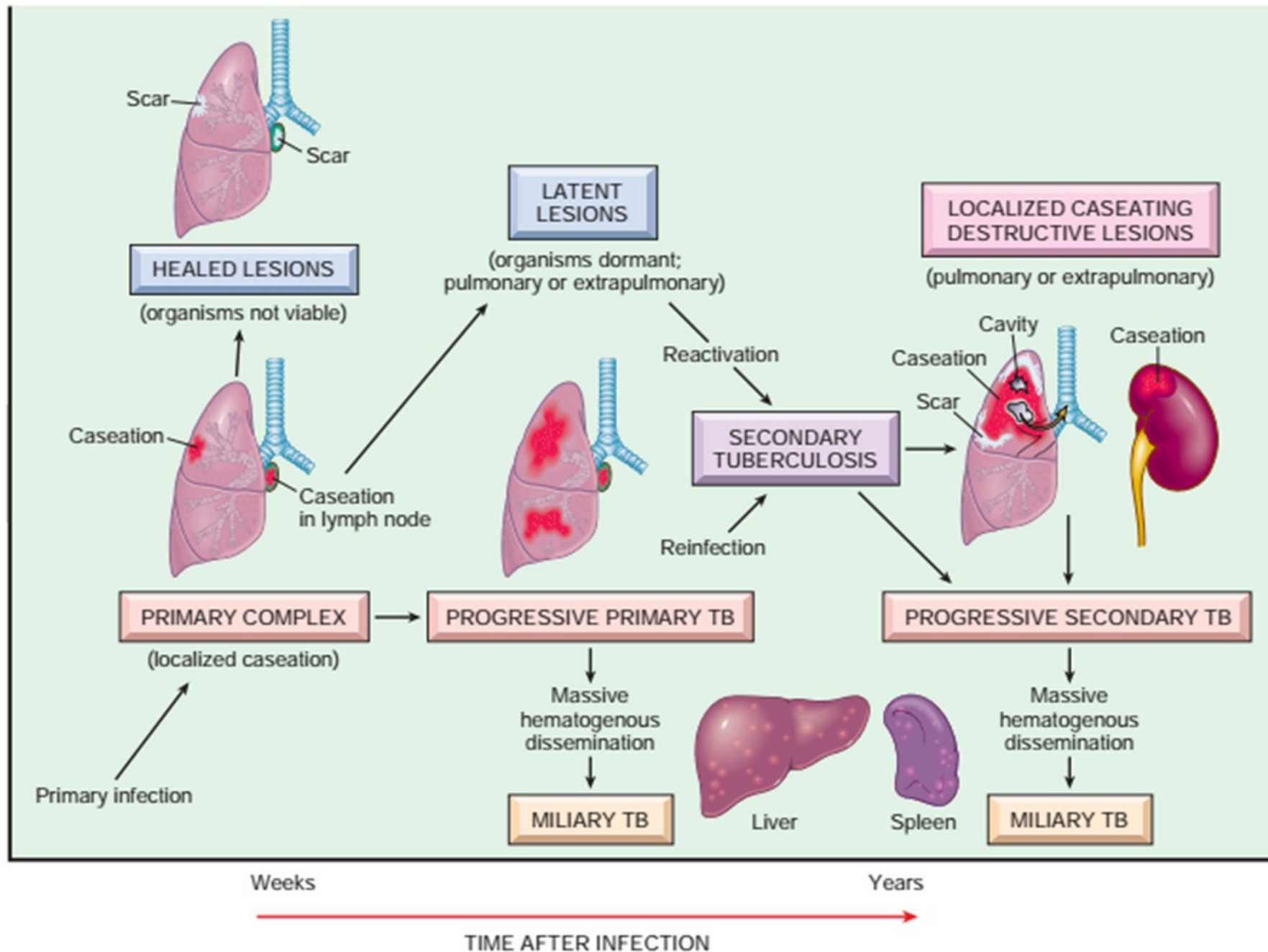
- ❖ Contracted by the drinking of **contaminated milk**
- ❖ In countries where milk is pasteurized
 - caused by the **swallowing of coughed-up infective material**
- ❖ Typically the organisms are seeded to mucosal lymphoid aggregates of the small and large bowel
 - **granulomatous inflammation** that can lead to ulceration of the overlying mucosa, particularly in the **ileum**.
- Healing creates **strictures**.

Clinical Course

- ❖ Localized secondary tuberculosis may be **asymptomatic**.
- ❖ Usually ***insidious*** in onset
- ❖ There is gradual development of both **systemic** and **localizing** symptoms.
- ❖ Systemic symptoms, probably related to **cytokines** released by activated macrophages (e.g., TNF and IL-1)
 - often appear early in the course and include **malaise, anorexia, weight loss, and fever**.

Ctd...

- ❖ The fever is **low grade** and **remittent**
- ❖ **Night sweats**
- ❖ Increasing amounts of **sputum**, at first mucoid and later purulent
- ❖ **Hemoptysis** is present in about **half** of all cases of pulmonary tuberculosis.
- ❖ **Pleuritic pain** may result from extension of the infection to the **pleural surfaces**.



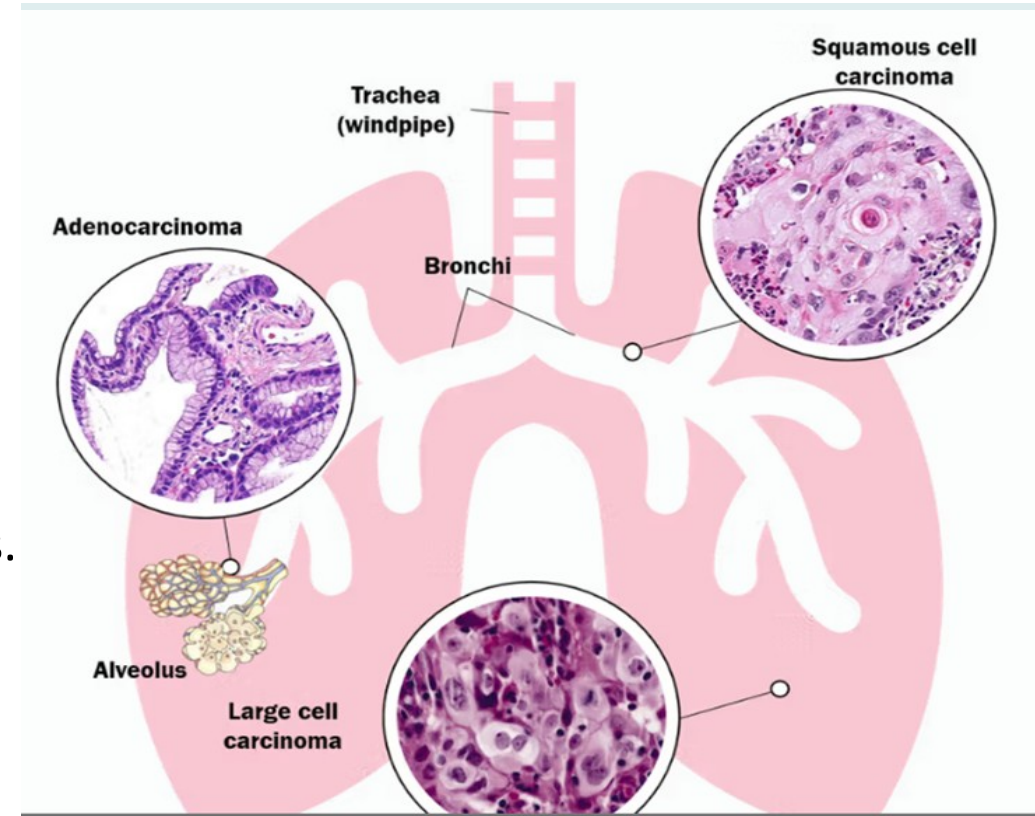
Pulmonary Neoplasia

- ❖ Pulmonary neoplasms are broadly classified into:
 - **Primary** Lung Tumors: Arising from the lung parenchyma, bronchi, or pleura.
 - **Metastatic** Tumors: The lungs are a common site for metastases from other organs.
- ❖ Lung cancer is the **leading cause** of cancer mortality worldwide for both men and women.
- ❖ Strongest Risk Factor: **Cigarette Smoking** (responsible for **~85%** of cases). Risk correlates with **pack-years**.
 - **Other Risk Factors:**
 - Radon gas exposure
 - Asbestos exposure (synergistic with smoking; associated with both lung carcinoma and mesothelioma)
 - Occupational exposures (arsenic, nickel, chromium), Air pollution
 - Family history and genetic predisposition
 - Pulmonary fibrosis (scar carcinoma)

Pulmonary Neoplasia

Classification:

- ❖ **Non-Small Cell Lung Carcinoma (NSCLC) (~85%):**
 - **Adenocarcinoma:** Most common type overall; arises from glandular cells, often **peripheral**. Associated with specific mutations (**EGFR, ALK**).
 - **Squamous Cell Carcinoma: Central**, arising from the bronchial epithelium; linked to keratin pearl formation and **cavitation**.
 - **Large Cell Carcinoma:** Undifferentiated, poor prognosis.
- ❖ **Small Cell Lung Carcinoma (SCLC) (~15%):**
 - Highly malignant, **central**, and strongly associated with smoking.
 - Arises from neuroendocrine cells
 - Often presents with **paraneoplastic syndromes**.



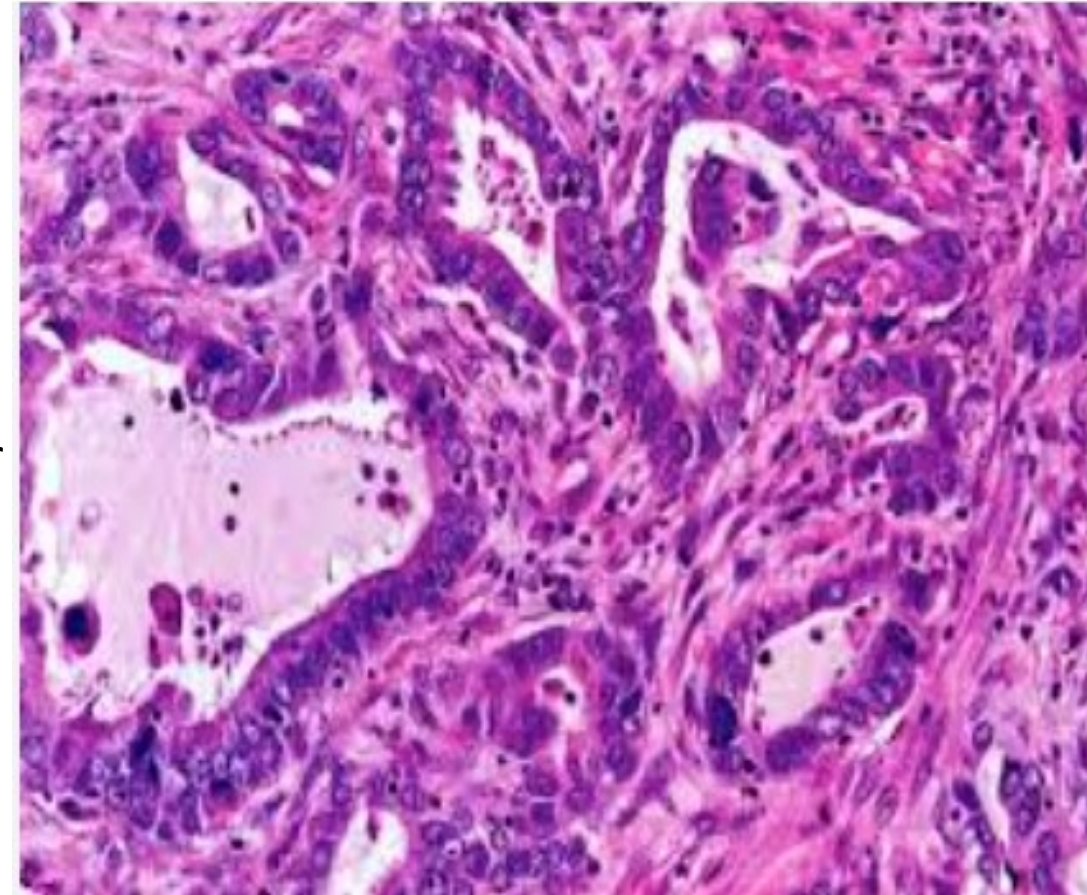
Non-Small Cell Lung Carcinoma (NSCLC)

A. Adenocarcinoma (Most Common type overall, ~40%)

- ❖ Origin: Bronchiolar or alveolar epithelium (Type II pneumocytes, Clara cells).
- ❖ Location: Often **peripheral**.
- ❖ Gross: Gray-white mass, often with stellate borders or pleural retraction.

Histopathology:

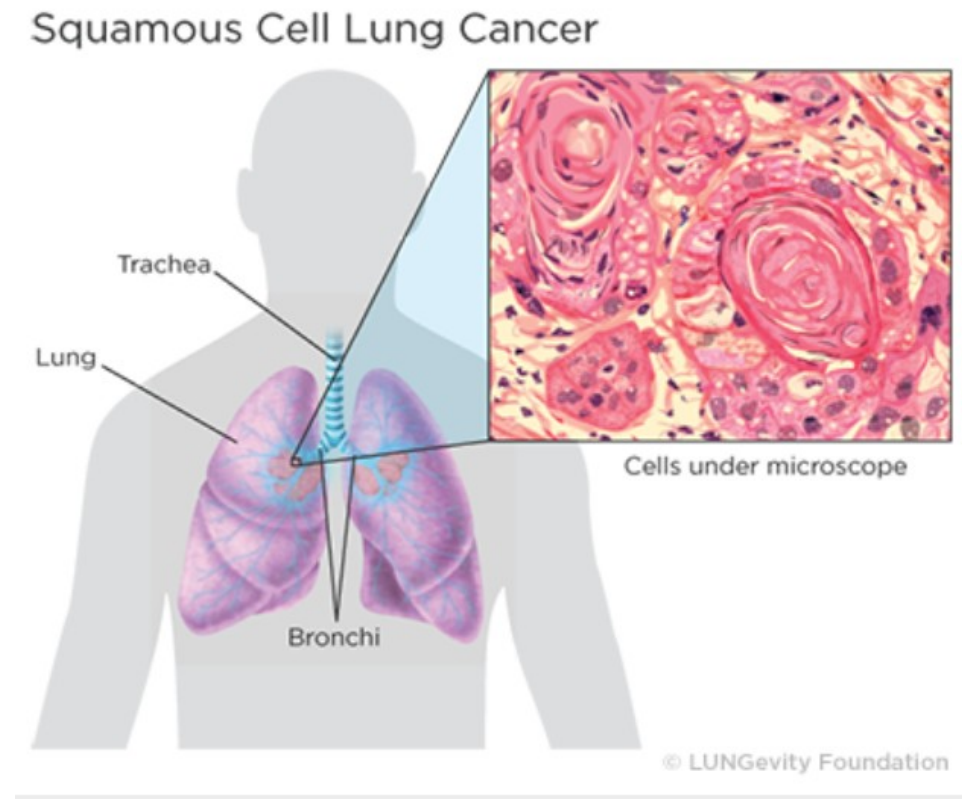
- ❖ **Lepidic Pattern:** Tumor cells grow along pre-existing alveolar structures (non-invasive). Pure lepidic growth = Adenocarcinoma in situ (AIS).
- ❖ **Acinar and Papillary Patterns:** Most common invasive patterns.
- ❖ **Micropapillary and Solid Patterns:** Associated with worse prognosis.
- ❖ **Cytology:** Tumor cells are columnar or cuboidal, with cytoplasmic mucin vacuoles.



Non-Small Cell Lung Carcinoma (NSCLC)

B. Squamous Cell Carcinoma (~20-30%)

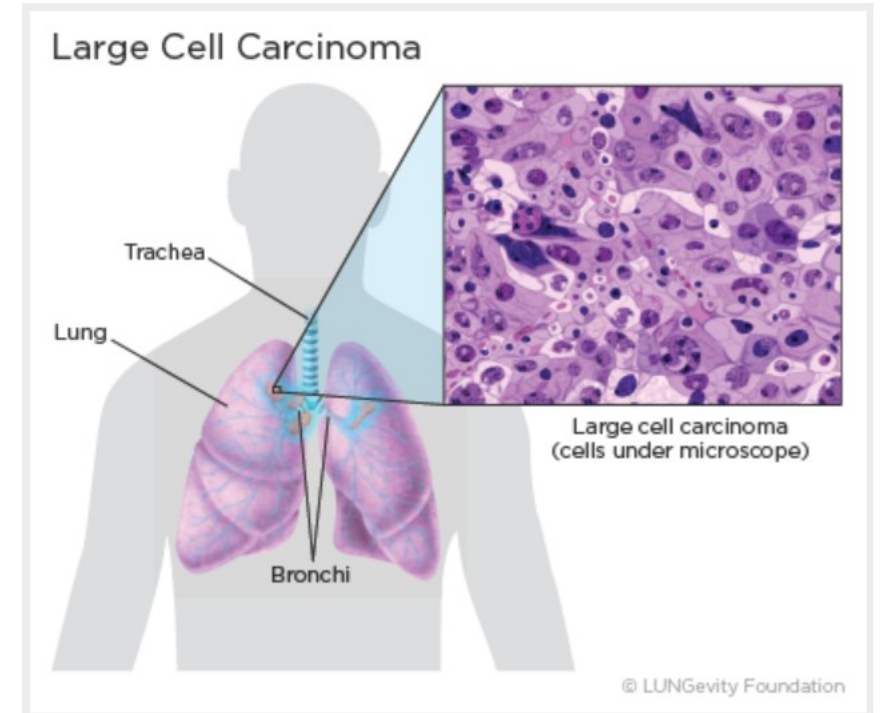
- ❖ **Origin:** Bronchial epithelium.
- ❖ **Location:** Often **central** (hilar), arising in a mainstem bronchus.
- ❖ **Gross:** Central, polypoid mass often causing airway obstruction; may **cavitate**.
- ❖ **Histopathology:**
 - Shows **keratinization** (individual cell keratinization, keratin pearls) and/or **intercellular bridges**.
 - Variants: Keratinizing, non-keratinizing, basaloid.



Non-Small Cell Lung Carcinoma (NSCLC)

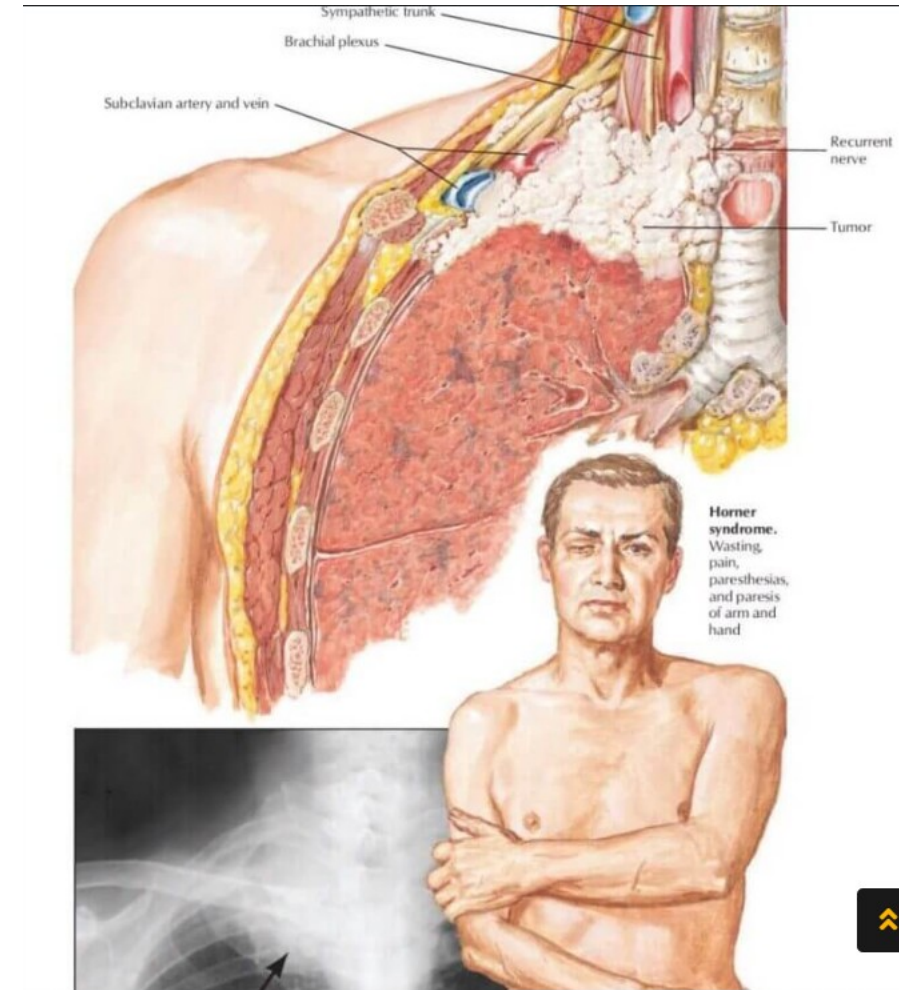
D. Large Cell Carcinoma (~5-10%)

- ❖ **Definition:** A **diagnosis of exclusion**.
- ❖ A poorly differentiated NSCLC
- ❖ Lacks light microscopic features of adenocarcinoma, squamous cell, or small cell carcinoma.
- ❖ **Histopathology:**
 - Sheets of large, polygonal cells with prominent nucleoli and abundant cytoplasm.
 - No specific differentiation.



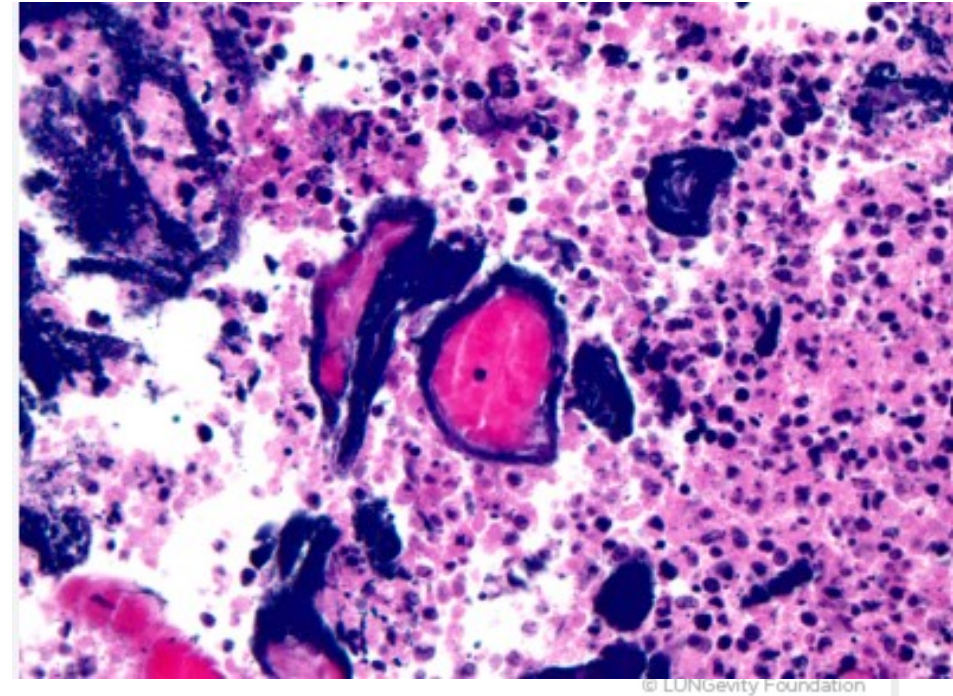
Pancoast Tumor

- ❖ Also known as a **superior sulcus tumor**
- ❖ Is a **carcinoma located in the lung's apex** in the **superior pulmonary sulcus**.
- ❖ Due to its **location**, it readily invades nearby structures, leading to a characteristic constellation of symptoms known as **Pancoast Syndrome**.
- ❖ **Histology:** Most commonly (over 95%) are **non-small cell lung carcinomas (NSCLC)**.
- ❖ **The Classic Triad of Symptoms:**
 - Severe, Unrelenting Shoulder and Arm Pain
 - Horner's Syndrome
 - Muscular Atrophy of the Hand Muscles:



C. Small Cell Lung Carcinoma (SCLC) (~15%)

- ❖ **Origin:** Neuroendocrine cells in the bronchial epithelium.
- ❖ **Location:** Almost always **central**.
- ❖ **Clinical:** **Highly aggressive, early metastases**, paraneoplastic syndromes common.
- ❖ Treated primarily with chemotherapy/radiation (rarely surgery).
- ❖ **Histopathology:**
 - Sheets of small, round, blue cells with:
 - "Salt-and-pepper" chromatin
 - High N:C ratio
 - Nuclear molding
 - Frequent mitoses and necrosis
 - Crush artifact (in small biopsies)



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Paraneoplastic Syndromes

❖ **Systemic** manifestations not caused by the tumor mass or metastases.

SCLC:

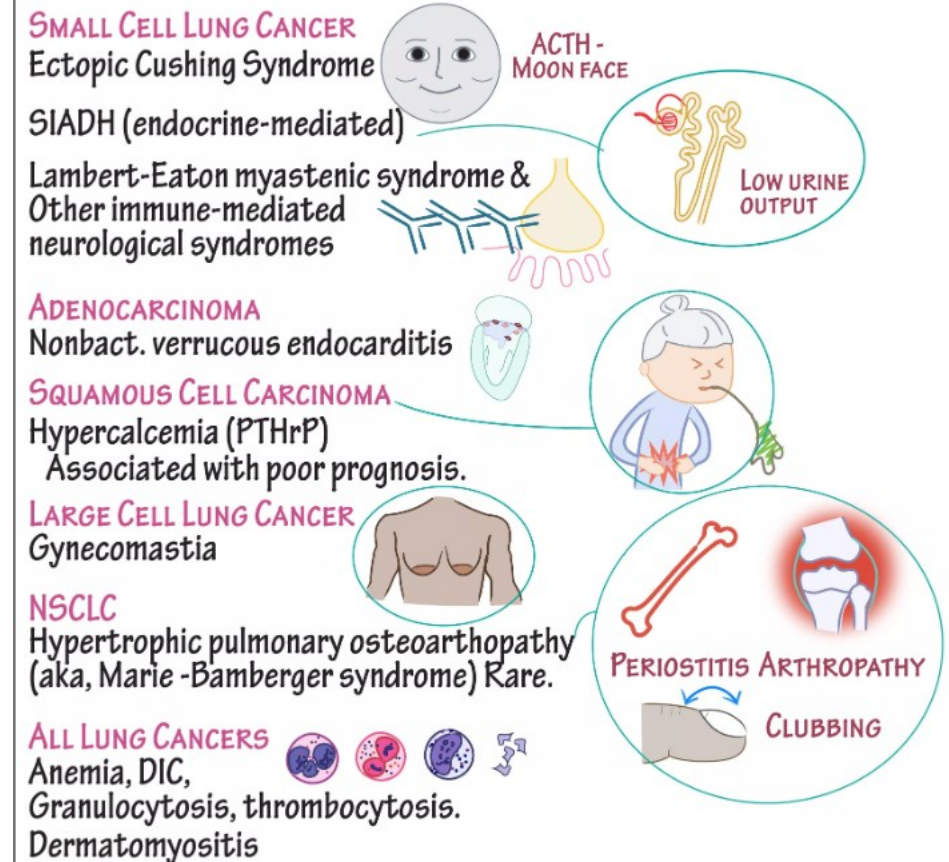
- ❖ **SIADH** (inappropriate ADH secretion)
- ❖ **Lambert-Eaton Myasthenic Syndrome**
- ❖ **Cushing's Syndrome** (ACTH production)

Squamous Cell Carcinoma:

- ❖ **Hypercalcemia** (PTHrP production)

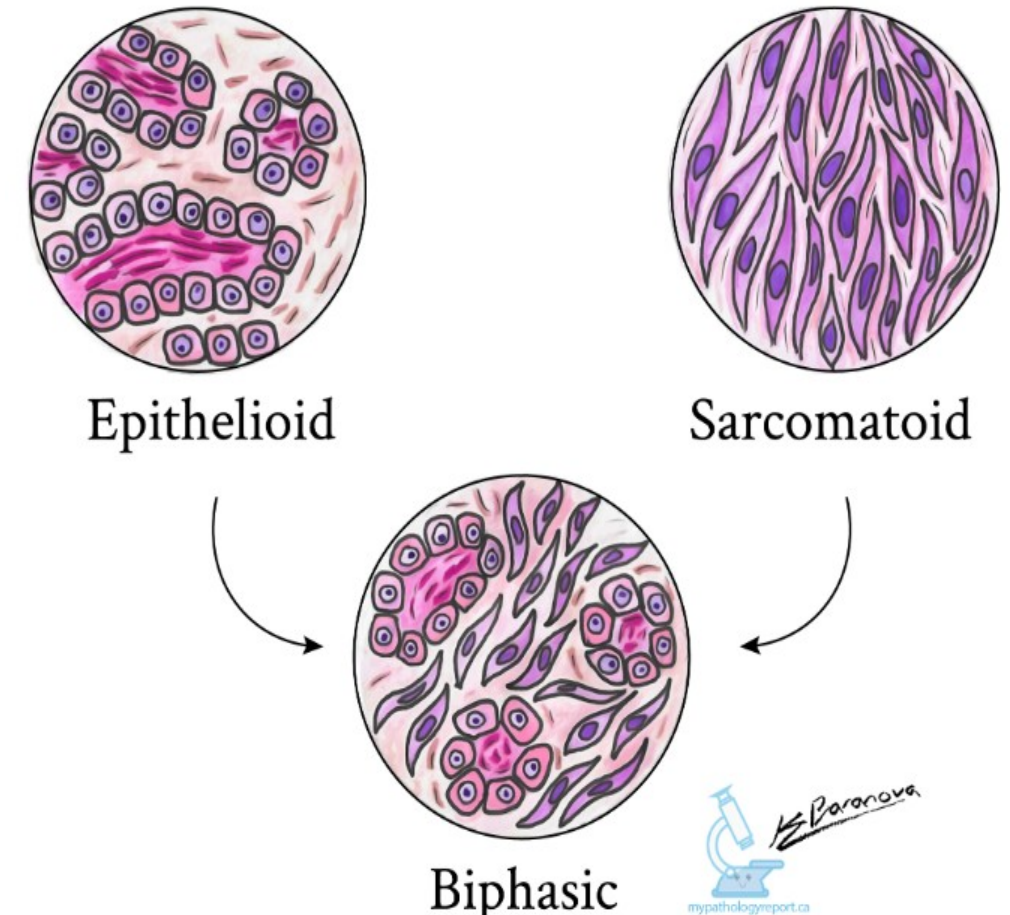
Adenocarcinoma:

- ❖ **Hypertrophic Pulmonary Osteoarthropathy (HPOA)** and **digital clubbing**
- ❖ **Migratory Thrombophlebitis** (Trousseau syndrome)



Mesothelioma

- ❖ **Origin:** Mesothelial cells of the pleura (rarely peritoneum).
- ❖ **Strongest Association: Asbestos exposure'**
- ❖ **Gross:** Diffuse, encasing growth of the lung ("pleural rind").
- ❖ **Histopathology:**
 - **Epithelioid** (most common): Tubulopapillary structures.
 - **Sarcomatoid:** Spindle cells.
 - **Biphasic:** Mixed.



Metastatic Tumors to the Lung

❖ **Common Primary Sites:** Breast, Colon, Kidney, Melanoma, Sarcomas.

❖ **Pathology:**

- Often multiple, well-circumscribed, spherical nodules ("cannonball" metastases on imaging).
- Histology resembles the primary tumor.

